

Tropical college of medicine



Obstetrics and gynecology Nursing II for 3rd year Bsc nursing student

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Content

I	• Abnormal pregnancy
II	• Antepartum hemorrhage
III	• Medical disease during pregnancy
IV	• Abnormal labor
V	• Operative delivery
VI	• Abnormal puerperium
VII	• Abnormalities of early pregnancy
VIII	• Menstrual disorder, infertility and menopause
IX	• Malformations of the female genital tract
X	• Benign and malignant tumors of genital tract

Unit- I:Abnormal pregnancy

Hyperemesis Gravidarum

Polyhydramnios

Hypertensive disorders of pregnancy

Introduction

- Morning sickness, complicates up to 70% of pregnancies.
- Typical onset is between 4 and 8 weeks' gestation with improvement before 16 weeks.
- 10 % to 25 % of women still experience symptoms at 20 to 22 weeks.
- Some women will have symptoms throughout the gestation.

Hyperemesis gravidarum

- Is severe form of nausea and vomiting during pregnancy resulting in dehydration and weight loss
- Frequently results in hospitalization
- The exact cause of hyperemesis gravidarum is not known.
- Possible etiology believed to be hCG, estradiol.
- The incidence of HG is only 0.3–1.5%(0.5-2%)

Risk factors for
hyperemesis
gravidarum

- ⑩ Younger age
- ⑩ Low prepregnancy body mass < 19.3 kg/m²
- ⑩ Female fetus
- ⑩ History of motion sickness or migraines
- ⑩ H. pylori infection ⇒ Iron deficiency
- ⑩ Primegravida
- ⑩ Multiple pregnancy
- ⑩ DM
- ⑩ Prior or family history of HEG
- ⑩ Trophoblastic disorder
- ⑩ Hyperthyroidism
- ⑩ Psychologic disorder
- ⑩ Overweight

Signs and symptoms

- Symptom and signs of complications
 - ⇒ Severe nausea and vomiting occurs >3x per day
 - ⇒ Loss of $\geq 5\%$ of pre-pregnancy body weight
 - ⇒ Dehydration, ketosis, and constipation
 - ⇒ Deficiency of Vitamin B1, B6, B12
 - ⇒ Alkalosis from loss of hydrochloric acid
 - ⇒ Hypokalemia

Investigations

- Common investigations include :
 - ⇒ Urinalysis, keton
 - ⇒ Stool Exam
 - ⇒ CBC, hematocrit levels, which are usually raised in HG.
 - ⇒ RBS
 - ⇒ Electrolytes- serum potassium, sodium, chloride
 - ⇒ Organ function test
 - ⇒ Thyroid function tests(↓TSH or elevated free thyroxine)
- Ultrasound scan

Outpatient management

- IV fluids

⇒ Infuse first liter over 1 -2 hours and then 1000 mls over 4 hours(i.e. 2 liters over 5 to 6 hours), followed by further assessment, including urine ketone testing.

⇒ Discharge the patient from outpatient care with PO medications and dietary advice. Or, transfer / admit for inpatient care.

Outpatient management...

- Dietary advise
 - ⇒ Advise the patient to avoid empty stomach
 - ⇒ Advise intake of small and frequent diet
 - ⇒ Restriction of coffee, spicy, odorous, high fat, acidic and very sweet foods
 - ⇒ Counsel on preferably taking protein dominant, salty (e.g. nuts), low fat, tasteless and dry snacks/meals.
 - ⇒ Encourage on fluid intake (better tolerated if cold, clear, and carbonated or sour)

Inpatient management

Indications for admission

Weight loss > 5% from pre-pregnancy

Ketonuria above +2

Electrolyte imbalance

Deranged renal and liver function tests

Persistent vomiting / failed OPD management

Inpatient management....

- Fluid management

- ⇒ Oral feeding withheld for 24 to 48 hours

- ⇒ Give 1 to 2 liters of NS or RL within 1 - 2 hours

- ⇒ Continue fluid repletion 1 -2 L over the next 2-3 hours

- ⇒ Maintenance fluid=1l of NS/RL + 4-6 40% glucose + 3 vitamin b complex for 8hours(43 drop/minute) until urine ketone free.

- ⇒ Anti-acid or H2 receptor blockers as needed for dyspepsia.

Inpatient management....

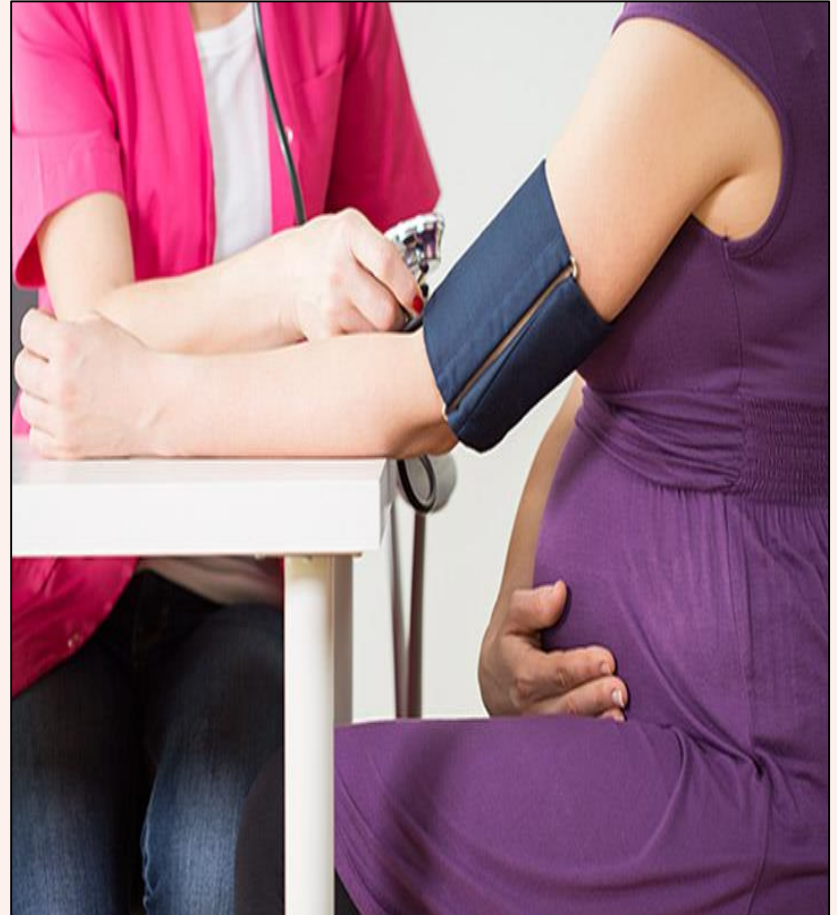
Antiemetic's	
First line	• Meclizine 25mg IV TID, or • Metoclopramide- 5–10 mg IV TID • Promethazine 5-10 mg IM every 6-8hrs
Second line	• Serotonin antagonists - ondansetron 4-8 mg IV or PO, TID
Third line	• Chlorpromazine 25mg IV or IM QID.

Hypertensive disorders

- The most common medical complications of pregnancy.
- Incidence is between 5% and 10%.
- Major cause of maternal and perinatal mortality and morbidity
- One of deadly triad:
 - 1.Hemorrhage
 - 2.Infection
 - 3.Hypertension

Hypertension in pregnancy

- A systolic blood pressure ≥ 140 mmHg, diastolic blood pressure ≥ 90 mmHg or both in two occasions taken 4 hours or more apart; or a single blood pressure recording of $\geq 160/110$ mmHg.



Hypertension in pregnancy classification...

1. Gestational hypertension

- ⇒ Hypertension without proteinuria
- ⇒ Hypertension without other signs of preeclampsia
- ⇒ Developing after 20 weeks of gestation in a previously normotensive woman.

Hypertension in pregnancy classification...

2. Preeclampsia eclampsia syndrome

- Preeclampsia

⇒ New onset of hypertension and proteinuria after 20 weeks of gestation in a previously normotensive woman.

- Eclampsia

⇒ Grand mal seizure or coma in a woman with preeclampsia.

Hypertension in pregnancy classification...

3. Chronic hypertension

- Hypertension that antedates pregnancy
- Is present before 20 weeks of gestation; or persists after 12 weeks postpartum.

4. Superimposed preeclampsia

- Superimposed pre-eclampsia without severe features
- Superimposed pre-eclampsia with severe features

Risk factors for preeclampsia

- First pregnancy
- Young or old age
- Multiple gestation
- History of hypertension
- Renal disease
- Diabetes
- Obesity
- Family history of pre-eclampsia

Classification of preeclampsia

Pre-eclampsia with severe features	Pre-eclampsia without severe features
• Blood pressure $\geq 160/110$ • Headache, blurred vision • Oliguria (<400 ml/24 hours) • Epigastric pain or pain in RUQ • Difficulty breathing • Low platelet count ($<100,000/\mu\text{l}$). • $\uparrow$ Liver enzymes more than 2x of upper limit of normal. • Serum creatinine higher than 1.1mg/dl	• Blood pressure $\geq 140/90$ • Proteinuria • No sign severe features

Clinical features of hypertension in pregnancy

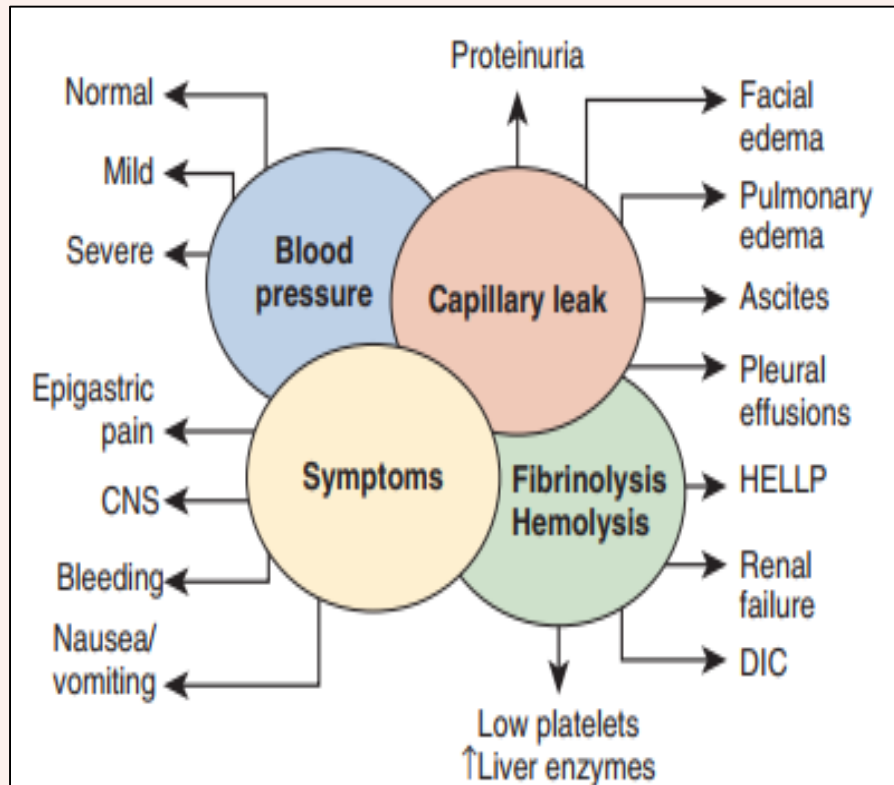


FIG 31-2 Maternal manifestations in preeclampsia. CNS, central nervous system; DIC, disseminated intravascular coagulation; HELLP, hemolysis, elevated liver enzymes, and low platelets.

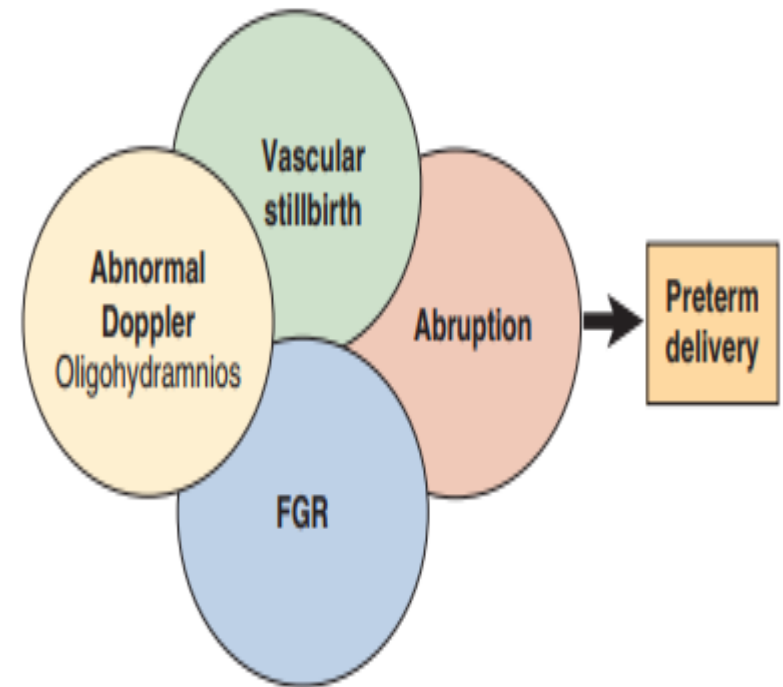


FIG 31-3 Fetal manifestations of preeclampsia. FGR, fetal growth restriction.

Clinical features of hypertension in pregnancy...



Fig. 17.1A: Marked pedal edema



Fig. 17.1B: Marked vulval edema in pre-eclampsia

Diagnosis

- Hypertension and proteinuria are the hallmark
- Severity features of preeclampsia
- Urine protein, CBC, liver enzymes, LDH and renal function test
- Ultrasound is used to monitor fetal growth and to assess fetal wellbeing.

Management of HDP

- Objectives
 - ⇒ Prevent eclampsia
 - ⇒ Preserve health of mother
 - ⇒ Deliver alive ,mature baby

Pre eclampsia with out severe features

- Gestational age less than 37weeks
 - ⇒ Monitor BP, fetal condition, CBC, OFT twice weekly(outpatient)
 - ⇒ Counsel danger signs associated with features of severe PE.
 - ⇒ Encourage the woman to eat a normal diet.
 - ⇒ Orient on fetal movement counting (kick chart) daily.
 - ⇒ Do not give anticonvulsants or antihypertensives
 - ⇒ Delivery at 37 completed weeks

Preeclampsia with severe features

General measures

Prevent convulsion

Control hypertension.

Delivery / expectant management

Preeclampsia with severe features...

- General Measures

- ⇒ Admit the patient urgently, preferably to the labor ward
- ⇒ Manage in left lateral position
- ⇒ Setup IV line & infuse maintenance fluids
- ⇒ Monitor urine output and maintain urine output at >30 ml/hr.
- ⇒ Monitor vital signs, FHB & reflexes
- ⇒ Auscultate the lung bases for fine crepitation.

Preeclampsia with severe features...

- Anticonvulsant therapy (seizure prophylaxis)
 - ⇒ Give Magnesium sulfate as shown in the box below.
 - ⇒ In all severe preeclamptic mothers during admission & continued during period of evaluation & observation for 24 hours.
 - ⇒ Diazepam: may be used as alternative, if MgSO₄ is not available

Preeclampsia with severe features...

- Control hypertension
 - ⇒ Start antihypertensives if the BP>160/110(one of it)
 - ⇒ Hydralazine or labetalol is the drug of choice for acute control.
 - ⇒ Maintain BP above the lower limits of normal(90-110mmhg)
 - ⇒ Maintaining
 - Systolic BP between 140 and 160 mm Hg
 - Diastolic BP between 90 and 105 mm Hg

Preeclampsia with severe features...

Hydralazine	• Give 5 mg IV slowly every 20 minutes until DBP<110 mmhg. • The maximum dose is 20 mg per 24
Labetalol	• PO 200 mg ⇒ Repeat dose after one hour until the treatment goal is achieved. ⇒ The maximum dose is 1200 mg in 24 hours. • 10 mg IV ⇒ If response is inadequate after 10 minutes, administer 20 mg IV. ⇒ The dose can be doubled to 40 mg and then 80 mg with 10-minute intervals until blood pressure is lowered below threshold. ⇒ The maximum total dose is 300 mg; then switch to oral treatment.
Nifedipine	• 10 mg orally. • Repeat dose after 30 minutes if response is inadequate until optimal BP. • The maximum total dose is 30 mg in the acute treatment setting. • For maintenance therapy 10-20 mg PO bid is given.
Alpha methyldopa	• Administer 250-750 mg every six to eight hours. • The maximum dose is 3000 mg per 24 hours.

Planning delivery

- Gestational age < 28 weeks $\Rightarrow$ Termination of pregnancy
- Gestational age $\geq$ 28 weeks and <34 weeks: expectant management if no contraindication
- Gestation after 37 Completed Weeks ,giving birth is recommended

Eclampsia

- Derived from a Greek word, meaning “like a flash of lightening”
- Pre-eclampsia when complicated with generalized tonic–clonic convulsions and/or coma.
- May occur quite abruptly, without any warning manifestations.
- 80% preceded by features of severe pre-eclampsia.

Eclampsia management

- General measures(IV fluid,position,suction,O2,spatula...)
- Control of convulsions(Mg₂So₄)
- BP control, stabilization of the condition of the mother & fetus
- Fluid balance.
- Delivery should take place within 12 hours of onset of convulsion
- Delivery & intrapartum/postpartum care

Anticonvulsant

MAGNESIUM SULFATE SCHEDULES FOR SEVERE PRE-ECLAMPSIA AND ECLAMPSIA

Loading dose

- Magnesium sulfate 20% solution, 4g IV over 5 minutes.
- Follow promptly with 10 gm of 50% magnesium sulfate solution, 5 gm in each buttock as deep IM injection with 1mL of 2% lidocaine in the same syringe. Ensure that aseptic technique is practiced when giving magnesium sulfate deep IM injection. Warn the woman that a feeling of warmth will be felt when magnesium sulfate is given.
- If convulsion recurs after 15 minutes, give 2 gm magnesium sulfate (20% solution) IV over 5 minutes.

Maintenance dose

- 5 gm magnesium sulfate (50% solution) + 1 mL lidocaine 2% IM every 4 hours into alternate buttocks.
- Continue treatment with magnesium sulfate for 24 hours after delivery or the last convulsion, whichever occurs last.

Complications hypertension in pregnancy

Maternal complications	Fetal complications
• Abruptio placentae • DIC • Acute renal failure • Hepatocellular injury, liver rupture • Brain damage • Transient blindness • Cardiorespiratory arrest • Aspiration pneumonitis • Acute pulmonary edema • Postpartum hemorrhage	• Fetal-growth restriction • Preterm birth • Fetal death

Polyhydramnios

- Is an abnormally increased amniotic fluid volume
- Definition
 - ⇒ Clinically when the volume exceeds 2000ml
 - ⇒ Sonography:
 - AFI ≥ 25 cm
 - Single vertical pocket (SVP) > 8 cm
- **Clinical states:** discomfort to mother

Cause polyhydramnios

1	Fetal anomalies	• Congenital malformations(20%) ⇒ Anencephaly: 50% ⇒ Open spina bifida ⇒ Esophageal atresia ⇒ Duodenal atresia ⇒ Facial clefts ⇒ Hydropes fetalis
2	Placenta	Chorioangioma
3	Multiple pregnancy	
4	Maternal:	• Poorly controlled DM, cardiac & renal diseases.
5	Idiopathic	~60%

Clinical presentation of polyhydramnios

- Severe dyspnea
- Palpitation
- Edema
- Markedly enlarged abdomen
- Fetal parts not felt easily
- FHB not heard distinctly

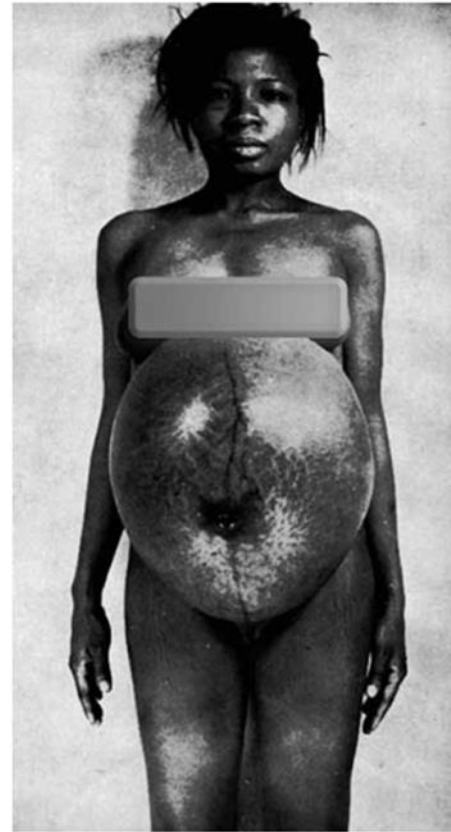


FIGURE 11-2 Severe hydramnios—5500 mL of amniotic fluid was measured at delivery.

Diagnosis of polyhydramnios

1	Clinical	• Uterine enlargement • Difficulty of breathing • Impossible to palpate any fetal parts
2	Sonography	• Differentiate hydramnios, ascites, ovarian cyst
3	Investigations:	• Ultrasound • Radiography • Blood: blood group, RBS, GTT • Amniotic fluid: AFP

Management of polyhydramnios

- Treatment options are tailored to the underlying cause.
- Poorly controlled DM $\Rightarrow$ control hyperglycemia
- During labor, manage as twin
 - $\Rightarrow$ Mild or moderate degrees: follow up
 - $\Rightarrow$ Severe
 - Dyspnea, abdominal pain, or difficult ambulation
 - Admit the women
 - Bed rest, diuretics, and water and salt restriction are ineffective.

Polyhydramnios complication

Maternal complications	Fetal complications
• Abortion • Risk of preeclampsia • Malpresentation • PROM • Preterm labor • Abruptio • Uterine inertia • Retained placenta • PPH • Sub involution	• Increased PNM 50% from ⇒ Prematurity ⇒ Congenital malformations • Cord prolapse

Oligohydramnios

- Is an abnormally decreased amount of amniotic fluid.
- Defined as
 - ⇒ AFI < 5 cm
 - ⇒ Amniotic fluid volume < 500ml
 - ⇒ Single vertical pocket < 2 cm
 - ⇒ Severe if MVP < 1cm

Conditions associated with oligohydramnios

Fetal	Maternal
• Chromosomal abnormalities • Congenital anomalies • Growth restriction • Fetal demise • Post term pregnancy • Ruptured membranes • Placenta abruption • Twin-to-twin transfusion • Renal agenesis	• Uteroplacental insufficiency • Hypertension • Preeclampsia • Dehydration-hypovolemia • Diabetes • Drugs ⇒ Prostaglandin synthase inhibitors ⇒ ACE inhibitors

Management

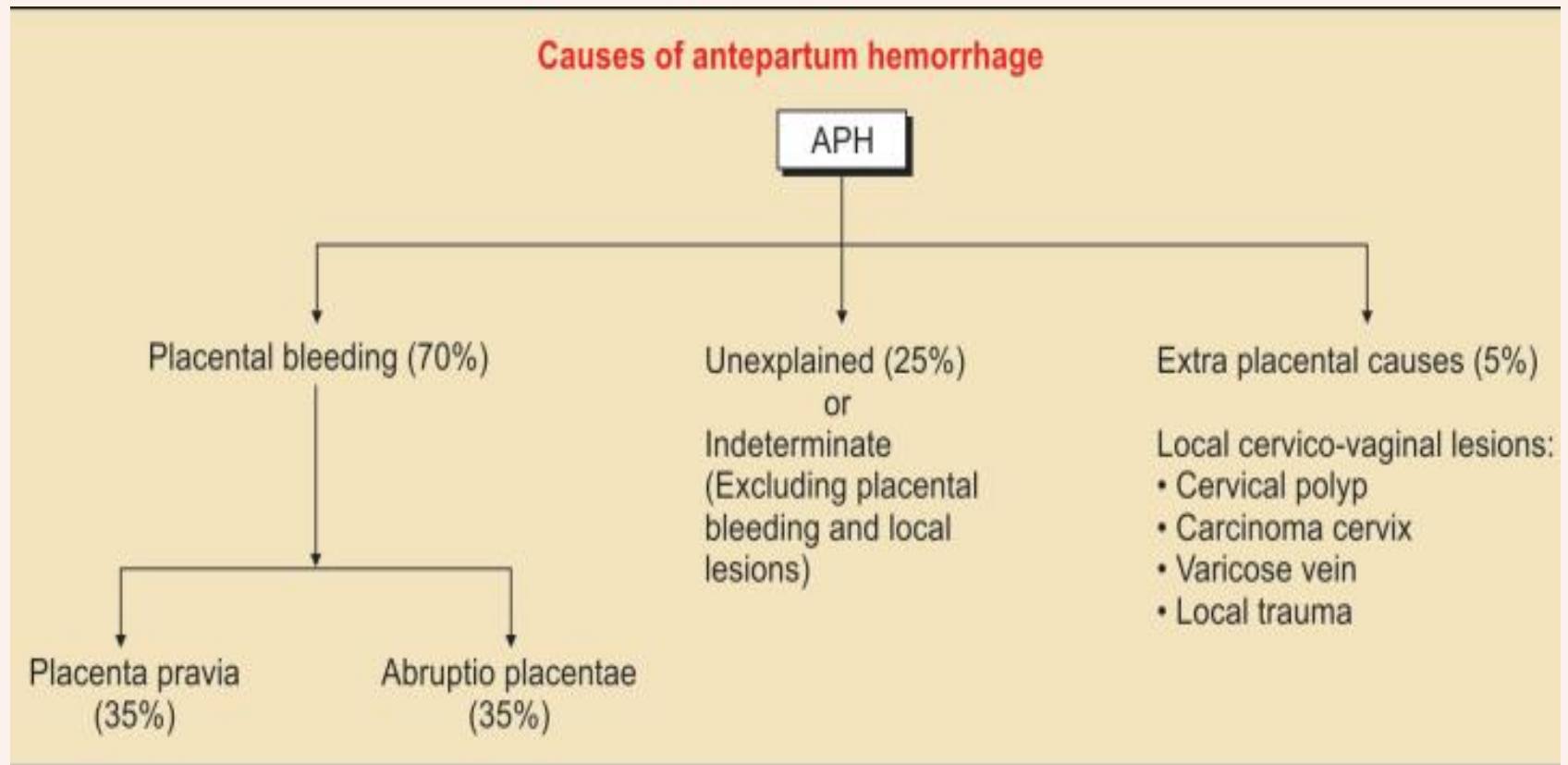
- Antepartum management may include maternal hydration.
- Amnioinfusion
 - ⇒ Is not the standard of care for other etiologies
 - ⇒ Is not generally recommended

Unit 2:Ante partum hemorrhage(APH)

- Is the occurrence of vaginal bleeding (VB) after 28th weeks of gestation and before delivery of the last fetus.
- Incidence is 1-4 % of pregnancy.



Causes APH



Placenta previa

- The implantation of the placenta over or near the internal os of the cervix (within 2cm) which diagnosed in 3rd TMP.
- 0.3 % to 0.4 % of all term pregnancies

Risk factors of placenta previa

1	Scarring or poorly vascularized endometrium	• Grandmultiparity • Advanced age(>35years) • Abortion
2	Need for increased placental surface area (large placenta)	• DM • Multiple gestation • Smoking • Male fetus
3	Prior placenta previa	• Increase 8x

Classification of placenta previa

1. Placenta previa(marginal PP, minor PP)

- Internal cervical OS is covered by placental tissue.

2. Low-lying placenta(marginal PP, minor PP)

- Placenta lies within 2 cm of the cervical OS
- Does not cover the cervical OS

Classification of placenta previa...



Low lying
placenta



Marginal
placenta



Partial
previa



Complete
previa

Clinical feature

- Painless, bright red, causeless, recurrent bleeding
- Uterus corresponds to the period of amenorrhea
- Uterus is soft and relaxed
- FHB is normal



Fig. 18.4: Labor in a woman with placenta previa complicated with prolapse of umbilical cord and footling. By Courtesy: Dr S Bhattacharya, Silchar

Investigations

- HCT
- Blood group and Rh
- Ultrasound: Fetal assessment, retroplacental clot and for localization of placenta

Treatment

- Resuscitate and stabilize on arrival and admit the patient.
- Assess maternal and fetal wellbeing.
- Prepare cross matched blood (at least 2 units).

Treatment...

- Expectant management indication

- ⇒ <37 weeks

- ⇒ Patient in stable condition

- ⇒ Reassuring fetal condition.

Treatment...

- Expectant management
 - ⇒ Give dexamethasone if GA < 37 weeks.
 - ⇒ Anti D 300µg IM if Rh negative and not sensitized.
 - ⇒ Closely monitor maternal and fetal conditions with APH chart.
 - ⇒ Prevent and treat anemia.

Treatment

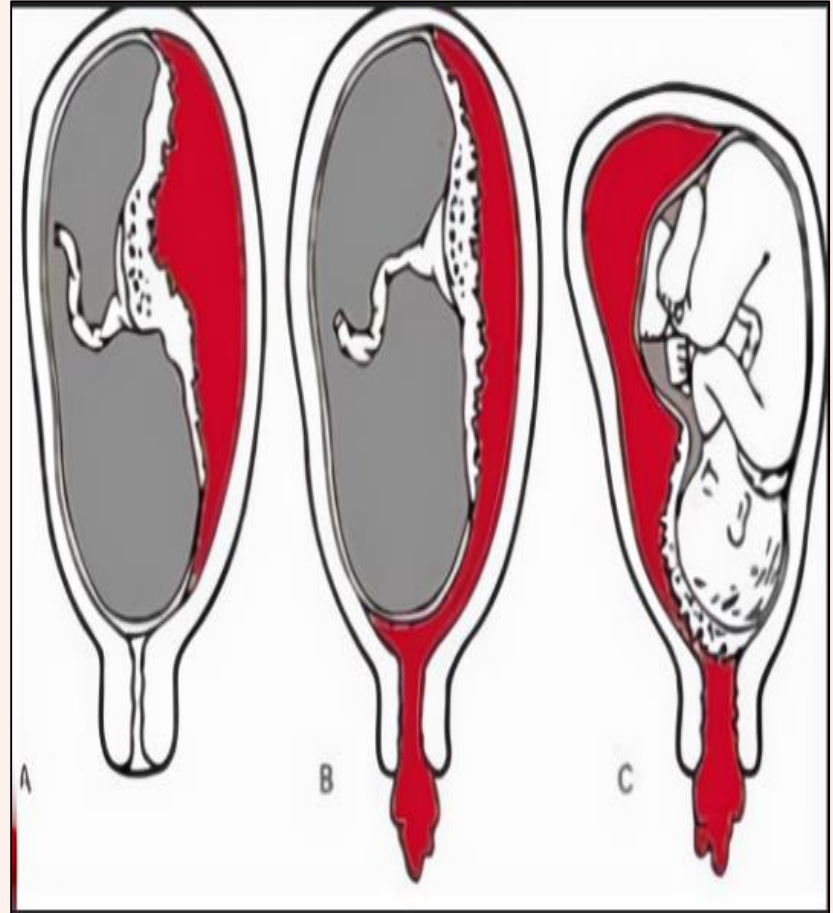
- Immediate delivery indication
 - ⇒ Gestational age is >37 weeks
 - ⇒ Deranged vital signs
 - ⇒ Heavy bleeding,
 - ⇒ NRFHRP, IUFD
 - ⇒ Lethal congenital anomaly of the fetus
 - ⇒ Established labor.

Treatment...

- Mode of delivery
 - ⇒ Vaginal delivery can be allowed for marginal placenta.
 - ⇒ Cesarean delivery:
 - Placenta previa(partial and complete)
 - Excessive bleeding
 - NRFHR

Abruptio placenta

- Premature separation of a normally implanted placenta after 28 weeks from the uterus before delivery of the last fetus.
- Classification.
 - A. Revealed(85%)
 - B. Concealed(15%)
 - C. Mixed



Risk factors

- Hypertensive disorders(30%)
- Multiparity
- Advanced age(>35yr)
- Metabolic abnormalities(eg, DM)
- Vascular disease(e.g, Systemic lupus erythematosus)
- Placental insufficiency
- Asthma

Classification of abruptio placenta.

Clinical presentation	Mild (Grade1)	Moderate (Grade 2)	Severe (Grade 3)
Degree of separation	<1/6 th	1/6 th -2/3 rd	>2/3 rd
Amount of blood	<400 mL	400-1000mL	>1000mL
Uterine irritability (pain/tenderness)	Normal / slightly increased	Increased	Tetanic, reactive,board like, tender on palpation
Fetal condition	No fetal heart abnormality	Abnormal / rarely	Fetal distress or death
Shock	Absent	Mild, postural hypotension	Severe and always present
BP	Normal	Postural hypotension	Reduced, recordable
PR	Normal	Increased	Feeble, fast
Fibrinogen level	Normal	Slightly decreased	Decreased
DIC	Absent	Absent	present

Clinical presentation

- Vaginal bleeding: menstrual-like (dark), totally concealed or the amount is less than the degree of the shock.
- Abdominal pain/ (uterine) tenderness.
- NRFHRP or absent fetal heart beat.
- Coagulation defect: bleeding (epistaxis, ecchymosis, petechiae).

Investigations

- HCT
- Blood group and Rh
- Coagulation profile: Platelet count, PT, PTT, fibrinogen or bedside clotting and bleeding tests
- Ultrasound: Fetal assessment, retroplacental clot and for exclusion of placenta previa

Treatment

- Resuscitate and stabilize on arrival and admit the patient.
- Assess maternal and fetal wellbeing.
- Prepare cross matched blood (at least 2 units).

Treatment...

- Expectant management indication

⇒ <37 weeks

⇒ Patient in stable condition

⇒ Reassuring fetal condition.

Treatment...

- Expectant management indication
 - ⇒ Give dexamethasone if GA < 37 weeks.
 - ⇒ Anti D 300μg IM if Rh negative and not sensitized.
 - ⇒ Closely monitor maternal and fetal conditions with APH chart.
 - ⇒ Prevent and treat anemia.

Treatment...

- Immediate delivery indication
 - ⇒ Gestational age is >37 weeks
 - ⇒ Estimated fetal weight is >2.5 Kg
 - ⇒ Deranged vital signs
 - ⇒ Heavy bleeding
 - ⇒ NRFHRP, IUFD, malformed fetus
 - ⇒ Established labor

Treatment...

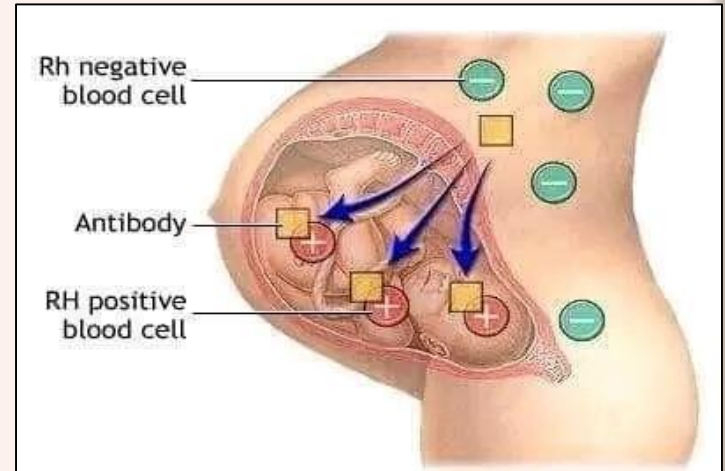
- Immediate delivery
 - ⇒ Mode of delivery: Vaginal delivery is preferred.
 - ⇒ Cervical ripening and induction of labor, amniotomy.
 - ⇒ Emergency cesarean section:
 - For severe bleeding endangering maternal life
 - NRFHR

Complications

- Post partum hemorrhage
- Hemorrhagic shock
- Organ failure (acute kidney injury, congestive heart failure).
- Disseminated coagulopathy
- Utero-placental insufficiency $\Rightarrow$ IUGR, fetal distress or IUFD

Red cell alloimmunization

- A process of exposure to foreign red cell antigens which results the production of anti-red cell antibodies
- Consequences known as hemolytic disease of the fetus and newborn (HDNF).



Nomenclature of RBC antigens

- Around 50 RBCs antigens are responsible for hemolysis.

1. Rh antigens

- D, C, c, E & e(CDE (Rhesus) Blood Group System)
- D antigen is more antigenic and commonly involved in Rh incompatibility

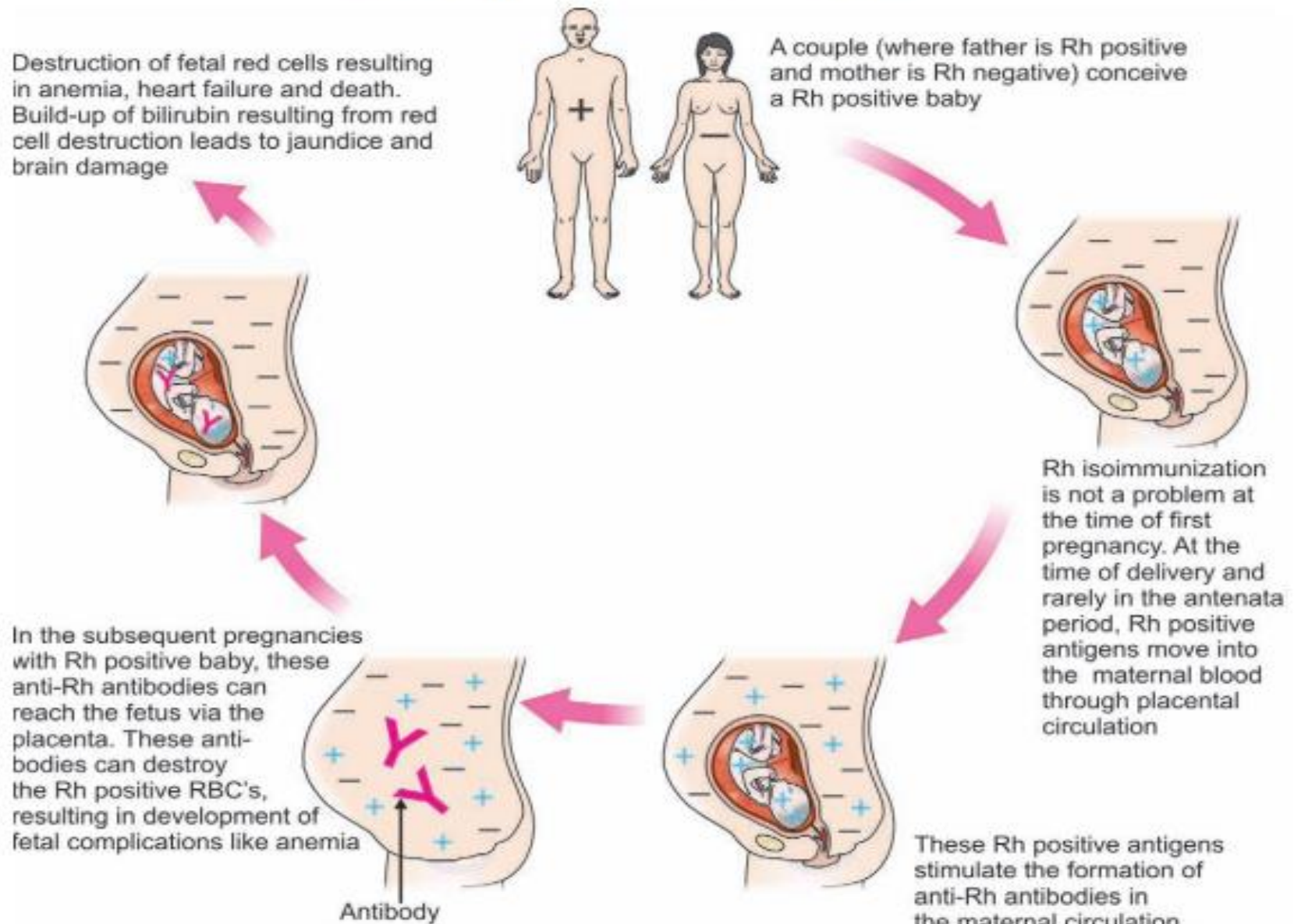
2. Kidd, Duffy, I, Lewis, etc

- Kidd(K1) is one of the common causes of fetal red cell hemolysis.

Risk factors for fetomaternal hemorrhage

Antepartum	Intrapartum/Intraoperative
⇒ Threatened abortion ⇒ APH ⇒ Multiple pregnancy ⇒ Fetal death ⇒ Abdominal trauma ⇒ Invasive obstetrical procedures ⇒ ECV	⇒ Abortion(induced or spontaneous) ⇒ Ectopic pregnancy ⇒ Instrumental delivery ⇒ C/S or hysterotomy ⇒ Manual removal of placenta ⇒ Postpartum curettage ⇒ GTD

Flow chart 6.1: Pathogenesis of Rh isoimmunization



Diagnostic tools

- Paternal blood group
- Maternal antibody titer
- Fetal blood group & antibody titer
- Ultrasonography
- Amniocentesis
- Doppler velocimetry (MCA)

Prevention of RHD HDFN

- If there is no evidence of anti-D alloimmunization in the RhD-negative woman, the patient should receive 300 µg of RhIG at 28 weeks.
- A dose of 50 µg of RhIG is effective until 13 weeks' GA
- AntiD within 72hours of post partum after checking fetal BG and Rh

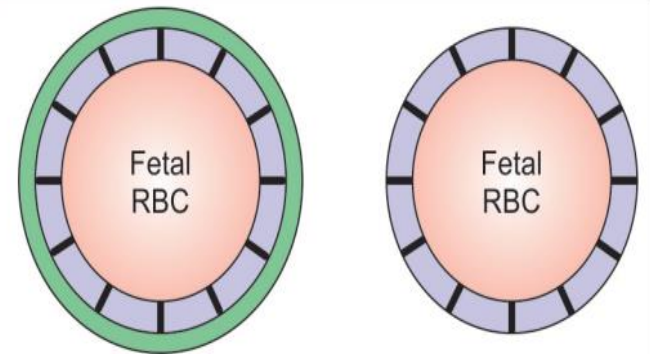


Fig. 22.2: Anti-D immunoglobulin injected into the mother surrounds the fetal RBC (thick green circle) preventing the antigen sites (spikes) to form antibody

Complications of sensitization

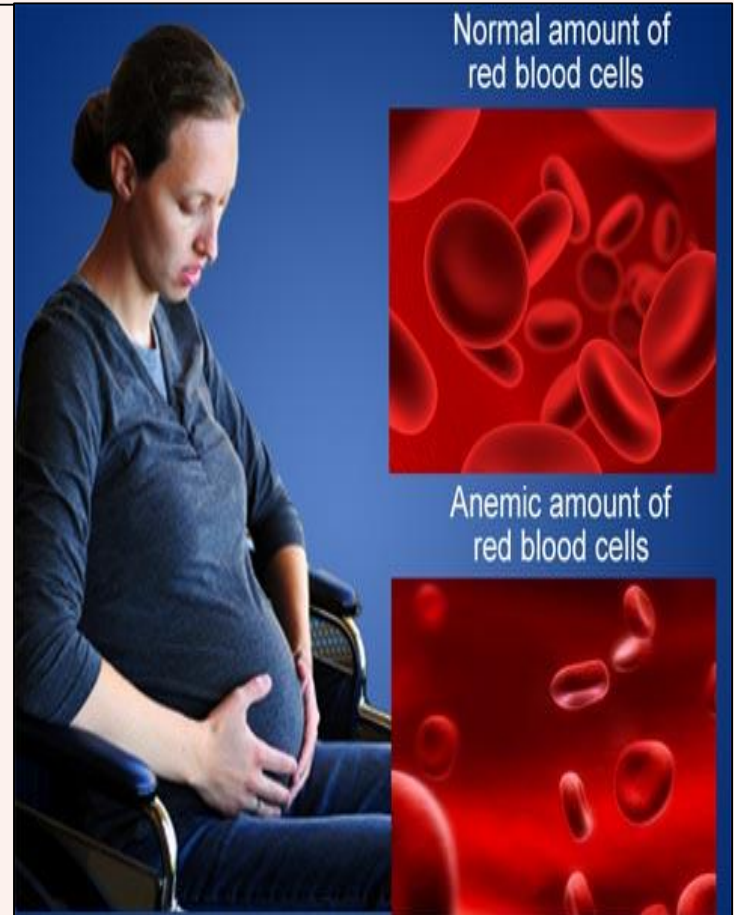
- Hemolytic anemia
- Hyperbilirubinemia (kernicterus)
- Hydrops fetalis
- Liver dysfunction

Unit-3:Diseases associated with pregnancy

- Anemia
- Assignment
 - ⇒ Diabetes mellitus
 - ⇒ Cardiac disease
 - ⇒ Bronchial asthma
 - ⇒ Malaria in pregnancy

Anemia in pregnancy

- Is decrease of RBC in blood
- The commonest disorder
- Anemia in pregnancy is a hemoglobin level of
 - ⇒ $<11\text{g/dl}$ in 1st TMP and 3rd TMP
 - ⇒ $<10.5\text{ g/dl}$ in 2nd TMP



Incidence

- The commonest medical complication of pregnancy.
- 50-65% of pregnant women develop anemia in low-resource countries
- Anemia is responsible for 20% of maternal deaths in the third world countries

Risk mother

- Multi-fetal gestations
- Poor socio-economic status
- Malnourished
- High parity
- Adolescent mothers
- Inter- pregnancy intervals less than two years
- Women with obstetric hemorrhage during current pregnancy

Cause of anemia during pregnancy

Common Causes—85% of Anemia

Physiologic anemia
Iron deficiency

Uncommon Causes

Folic acid deficiency
Vitamin B₁₂ deficiency(due to the rapid increase in bariatric surgery)
Hemoglobinopathies

- Sickle cell disease
- Hemoglobin SC
- β -Thalassemia minor

Bariatric surgery
Gastrointestinal bleeding

Rare Causes

Hemoglobinopathies

- β -Thalassemia major
- α -Thalassemia

Syndromes of chronic hemolysis

- Hereditary spherocytosis
- Paroxysmal nocturnal hemoglobinuria

Hematologic malignancy

Symptoms of Anemia in Pregnancy



Excessive tiredness



Weakness



Headaches



Dizziness



Chest pain



Shortness of breath



A rapid or irregular heartbeat



Pale skin



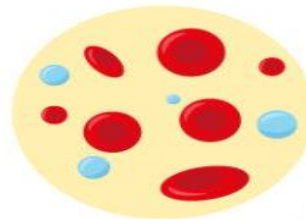
A numb or cold feeling in hands and feet



A low body temperature



Irritability



Diagnostic approach to anemia in pregnancy

Nurse role in the assessment of women with anemia in pregnancy

- **History taking:** hx of any of the predisposing factors.
- **symptoms of anemia :**
 - C.V.S→ palpitation.
 - C.N.S→ headache , visual disturbance , weakness , fatigability , drowsiness.
 - R.S→ breathlessness.
 - G.I.S→ anorexia , nausea , vomiting.
 - G.U.S→ loss of libido.

- History and examination
- Routine Hemoglobin check at first ANC visit
- If anemia is diagnosed
 - ⇒ Intestinal parasites stool exam
 - ⇒ Malaria by blood film

Complications of anemia during pregnancy

Fetus

- Neural Tube defects (esp. folate def.)
- Miscarriage
- OIUGR/ Low birth weight
- Prematurity
- Anemia in infancy
- IUFD

Mother

- Susceptibility to infection
- Heart decompensation and Heart failure
- Preterm labour and Preterm delivery
- Post-partum hemorrhage
- Mental lassitude & Loss of working hours
- Death

Management of anemia during pregnancy

- Depends on the specific cause
- Iron rich diet nutritional anemia
- Treat the cause (Hookworm, malaria ...)
- Elemental Iron from 120-200mg daily
- Blood transfusion for severe anemia (Hgb <7gm/dl)

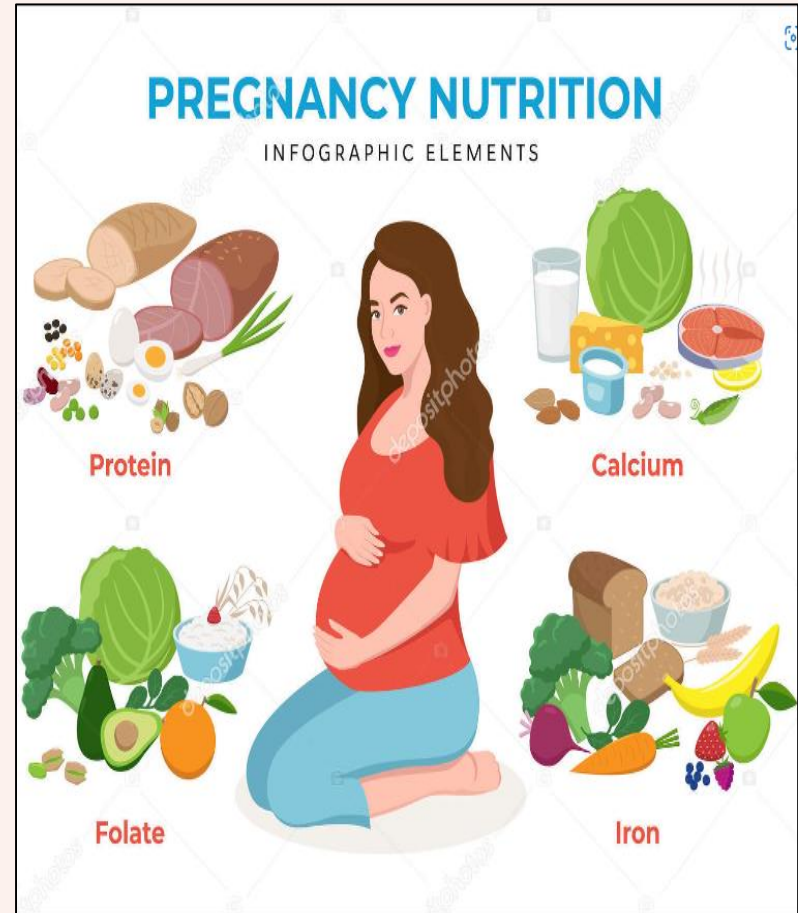
TABLE 42-2

ELEMENTAL IRON AVAILABLE FROM COMMON
GENERIC IRON PREPARATIONS

PREPARATION	ELEMENTAL IRON (mg)
Ferrous gluconate 325 mg	37-39
Ferrous sulfate 325 mg	60-65
Ferrous fumarate 325 mg	107

Prevention

- Adequate iron rich diet
 - Prevent infection
 - Prophylaxis iron
- ⇒ One 325 mg tablet of ferrous sulfate daily
- Wear shoes
 - Interpregnancy interval(2years)



Assignment(30%)

- Discuss about medical disorder of pregnancy

1. Gestational Diabetes
2. Heart disease
3. Malaria in pregnancy

- Content of assignment

- ⇒ Objective
- ⇒ Definition
- ⇒ Incidence
- ⇒ Risk factor
- ⇒ Etiology
- ⇒ Clinical presentation
- ⇒ Investigation
- ⇒ Diagnosis
- ⇒ Management
- ⇒ Complication on pregnancy
- ⇒ Reference

Unit- 4:Abnormal labor

Mal position

Mal presentation

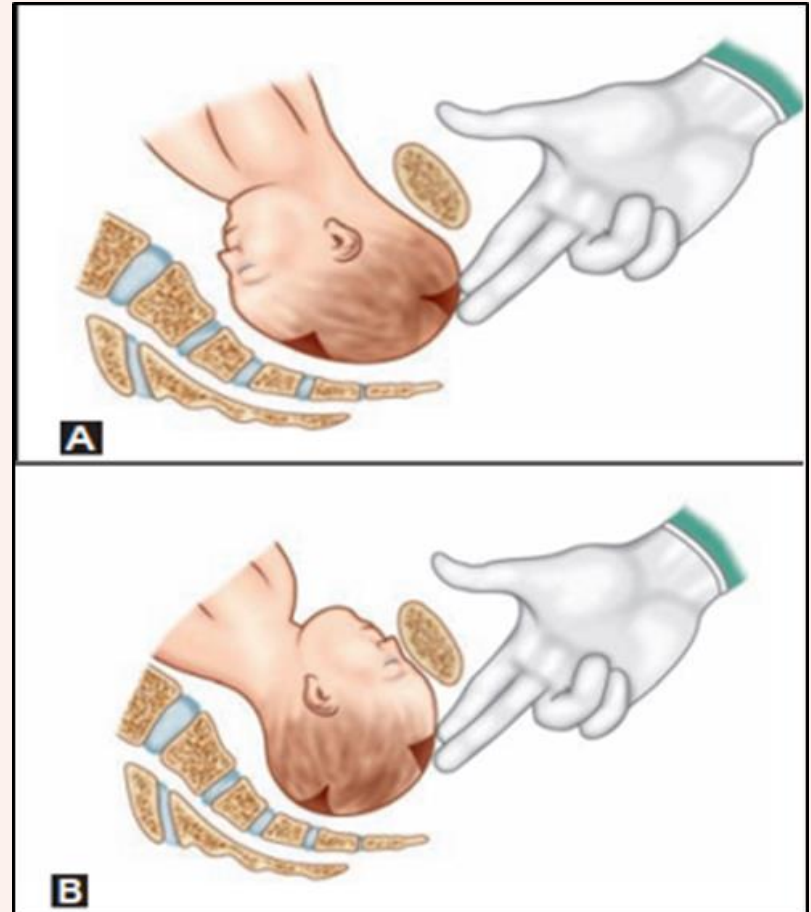
Cephalpelvic disproportion(CPD)and obstructed labor

Malposition and malpresentation

- Denominator

⇒ Part of the fetus which determines the presenting part

1. Vertex-Occiput
2. Breech-Sacrum
3. Face—mentum
4. Brow-sinciput



Fetal position

- Is the relationship between the denominator of the presentation to the six areas of the mother pelvis.



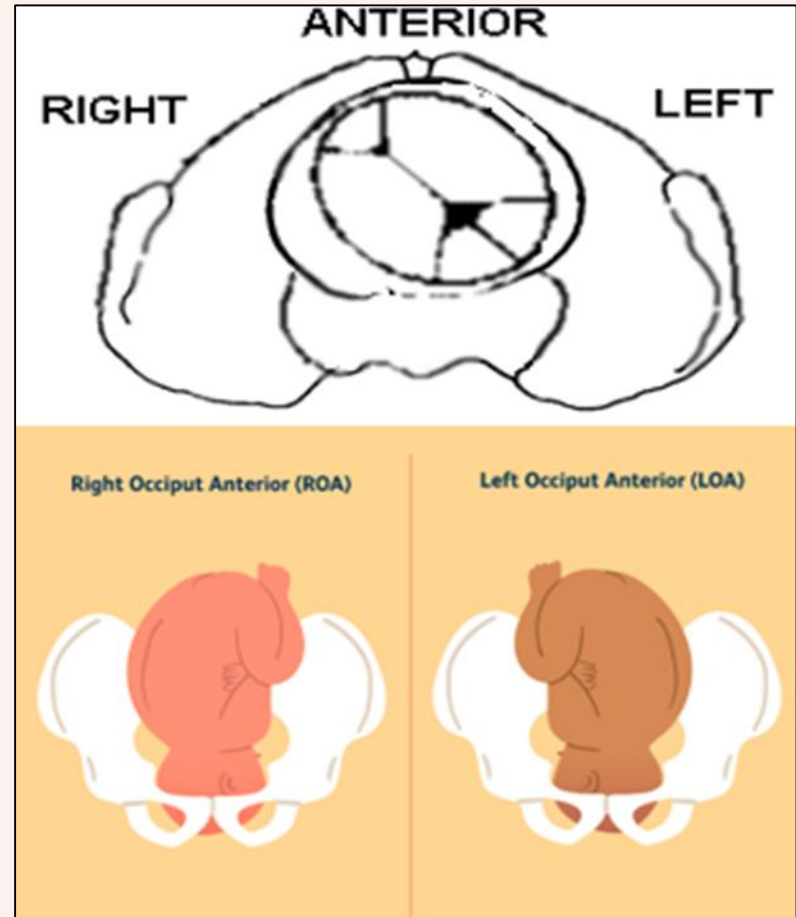
Fetal position...

- The normal fetal position is occiputo anterior

⇒ Direct

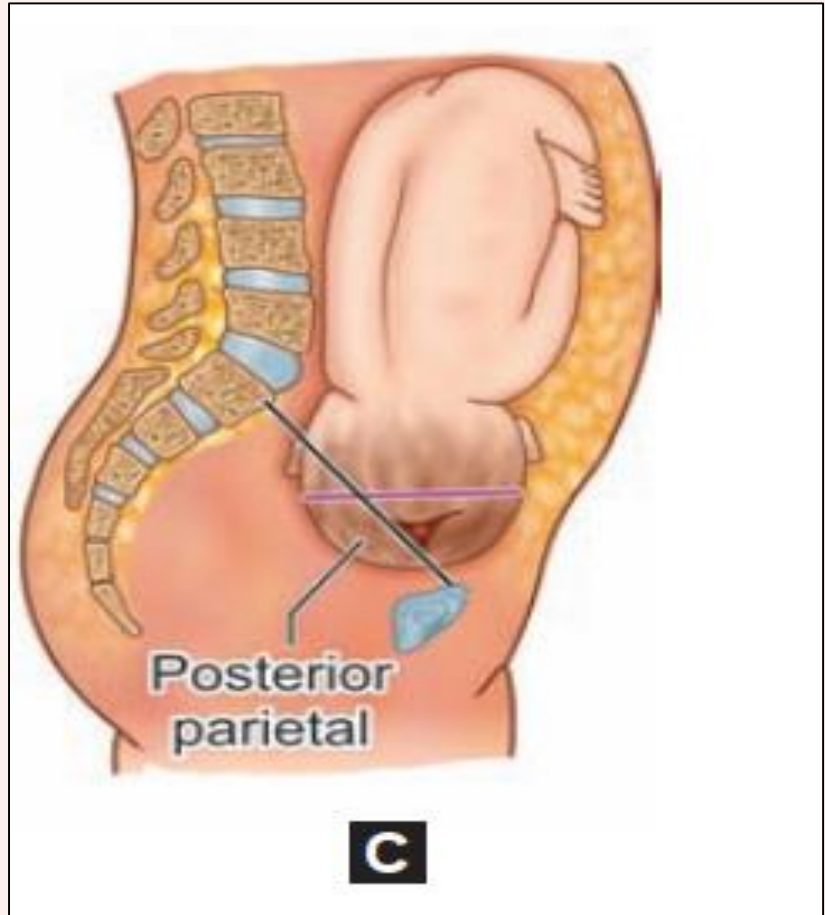
⇒ Right

⇒ Left

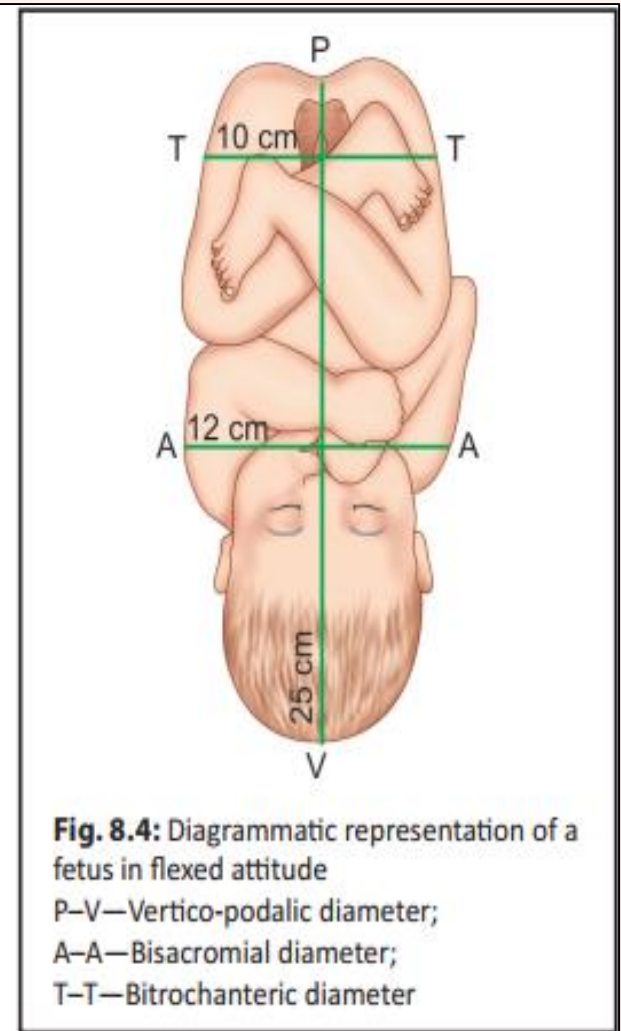
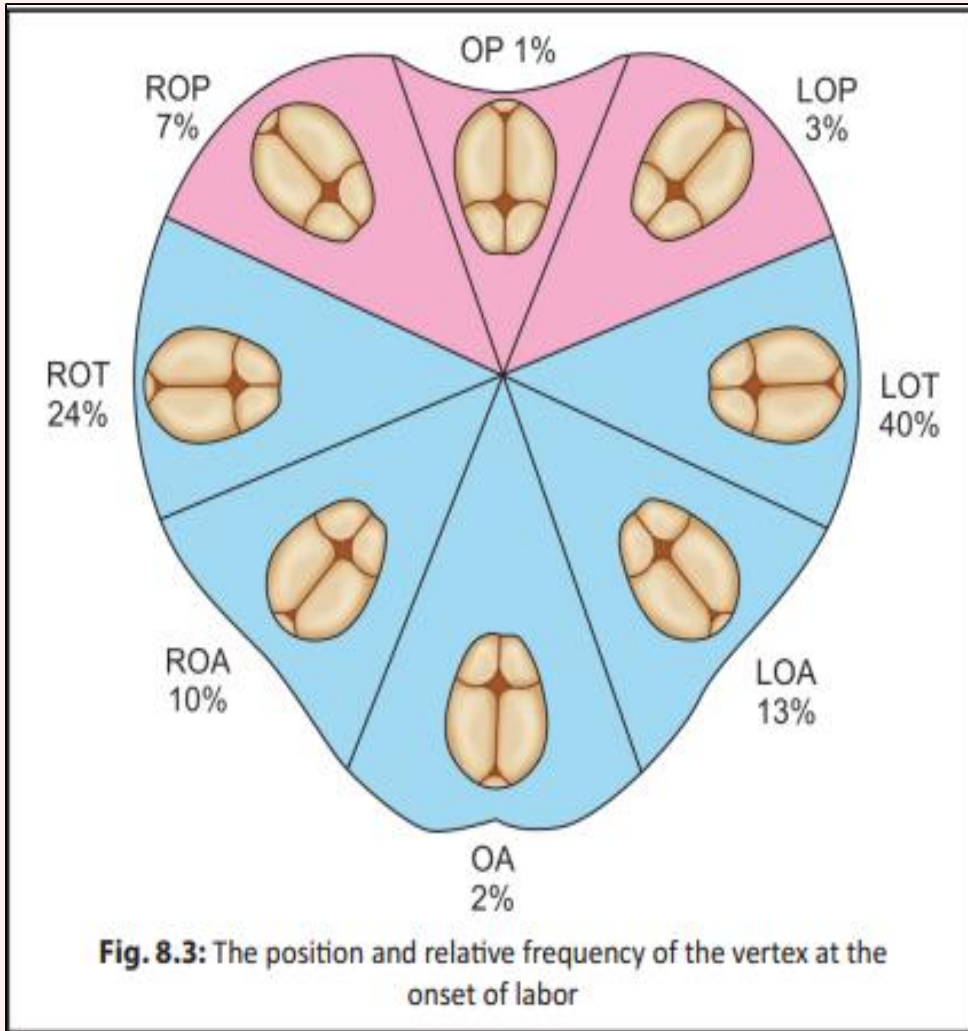


Malposition

- Malposition are any position of the vertex other than flexed occipito-anterior one
- Include
 - ⇒ Occiput posterior
 - ⇒ Persistent occiput transverse



Malposition...

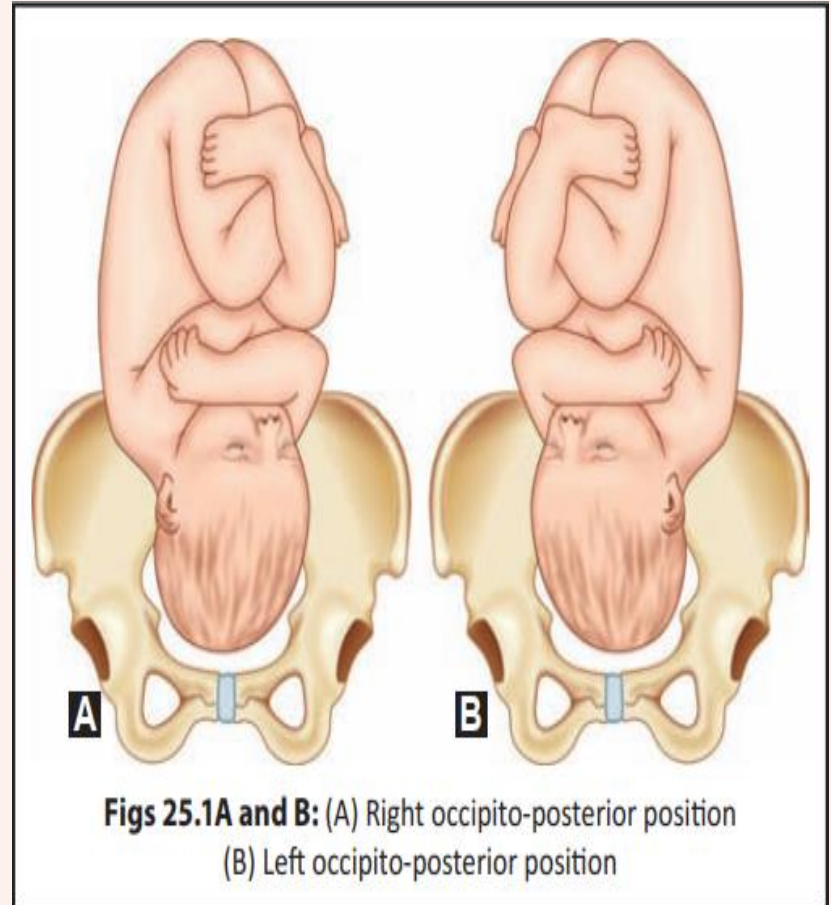


Predisposing factors

Maternal	Fetal and placental
• Contracted pelvis • Pelvic tumors: uterine myomas, ovarian tumors etc • Uterine anomalies: bicornuate uterus, uterine septum etc • High parity	• Prematurity • Fetal anomaly (e.g. hydrocephalus, anencephalus) • Polyhydramnios / oligohydramnios • Multiple pregnancy • Placenta previa

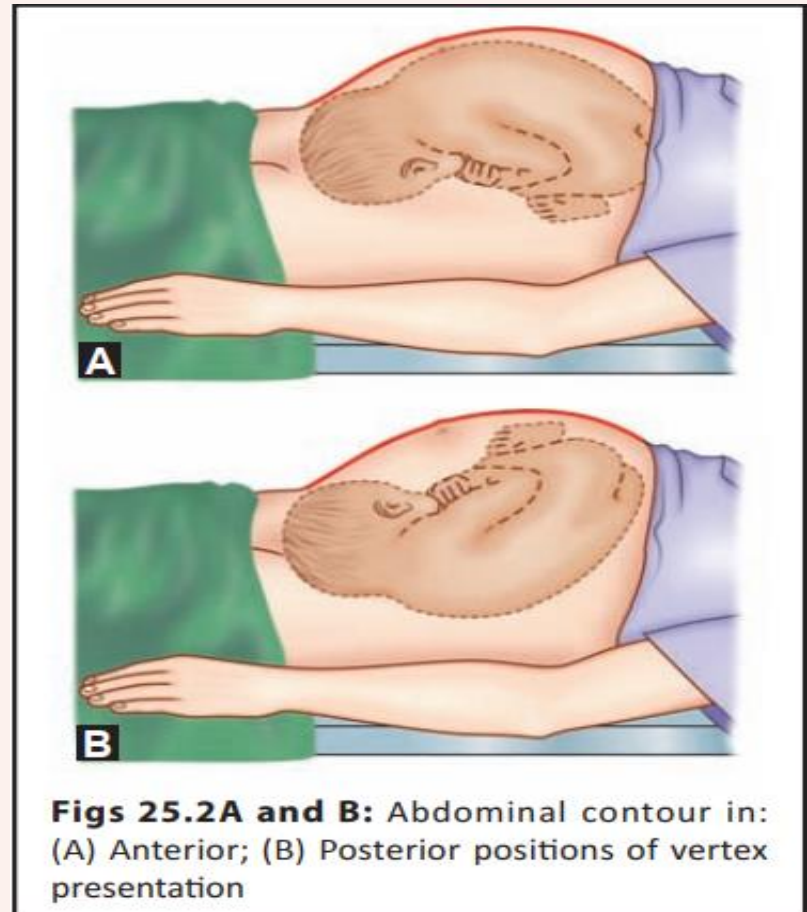
Occiput posterior position

- Is when the fetal occiput is posterior in relation to the maternal pelvis



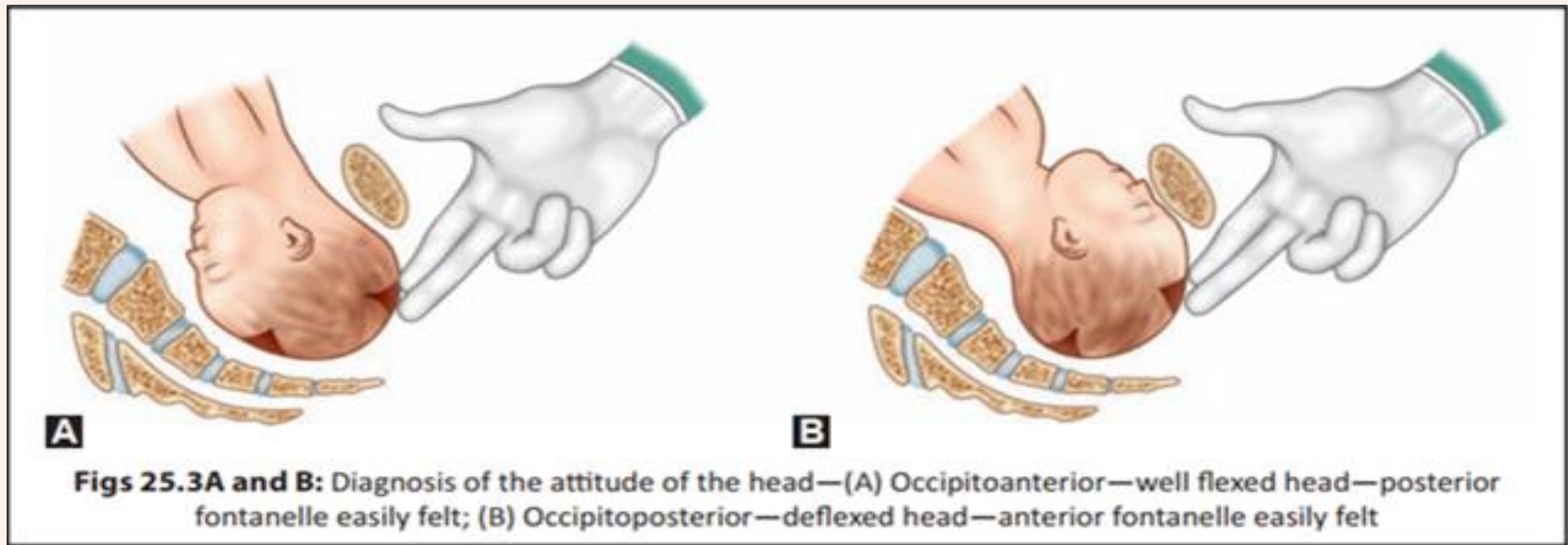
Diagnosis

- Suggestive abdominal findings
 - ⇒ Flattened lower part of abdomen.
 - ⇒ Anteriorly palpable fetal limbs.
 - ⇒ Fetal heart heard in the flank.



Diagnosis

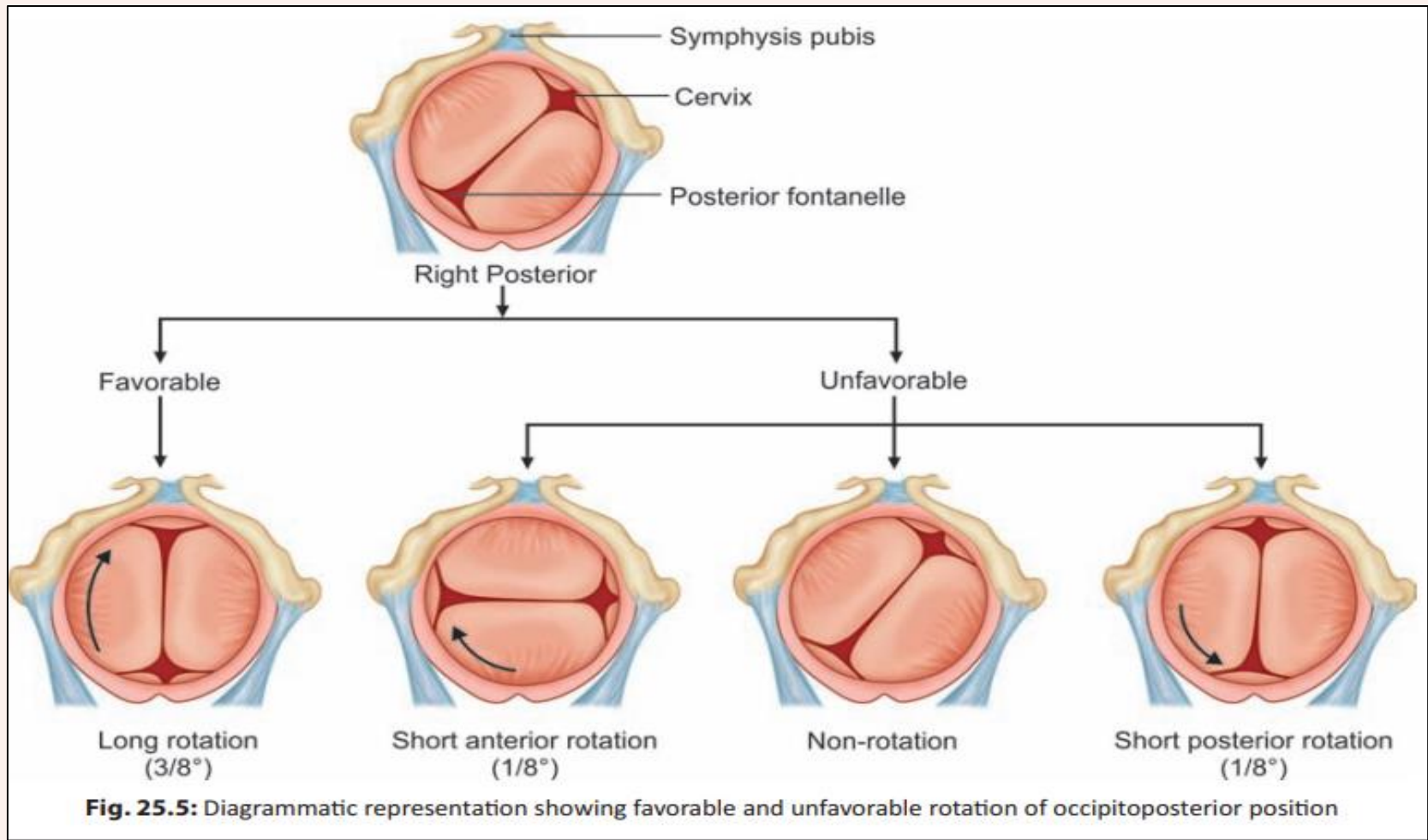
- On vaginal examination
 - ⇒ Posterior fontanelle towards the sacrum.
 - ⇒ Anterior fontanelle felt anteriorly if neck is flexed.



Management

- Rotation to occiput anterior, expect vaginal delivery
- Incomplete rotation leading to occipito-transverse position:
 - ⇒ Expect vaginal delivery
 - ⇒ Vacuum delivery(below station +2)
 - ⇒ Deliver by cesarean section

Management

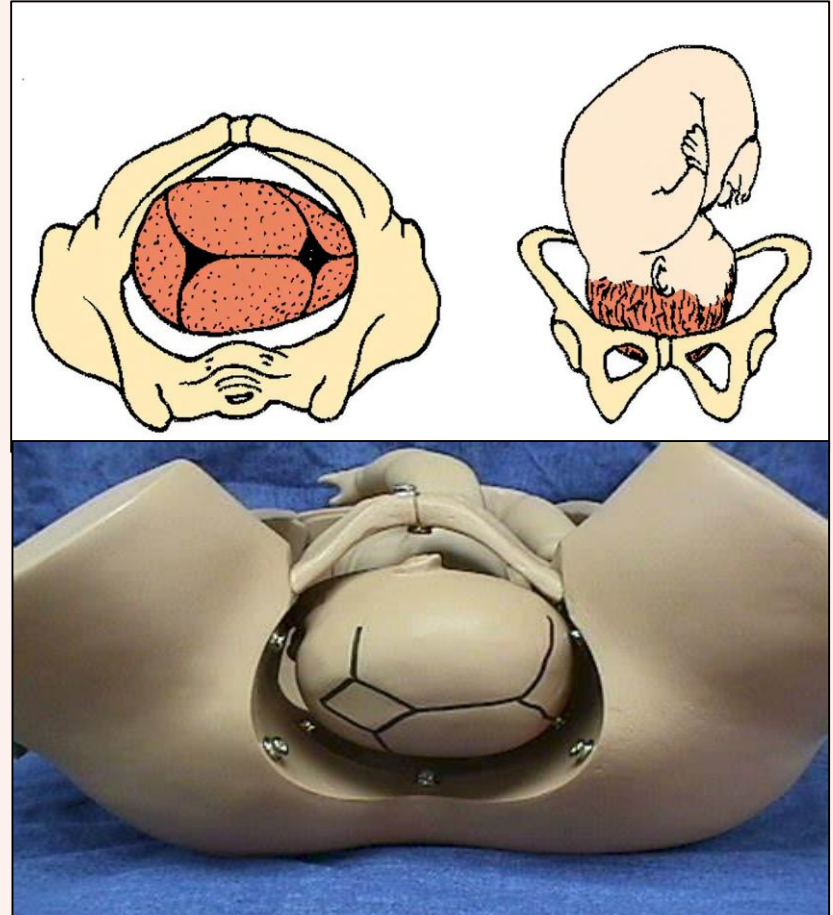


Management

- If no rotation
 - ⇒ Expect vaginal delivery
 - ⇒ Augment with oxytocin
 - ⇒ Vacuum delivery
 - ⇒ Deliver by cesarean section

Persistent occiput transverse position

- Is an occiput transverse position that is maintained for an hour or more in the second stage of labor.



Causes and risk factors

- Inadequate power (contraction and poor pushing)
- Platypelloid and android pelvis.
- Fetal head in long occipito-frontal diameter

Diagnosis

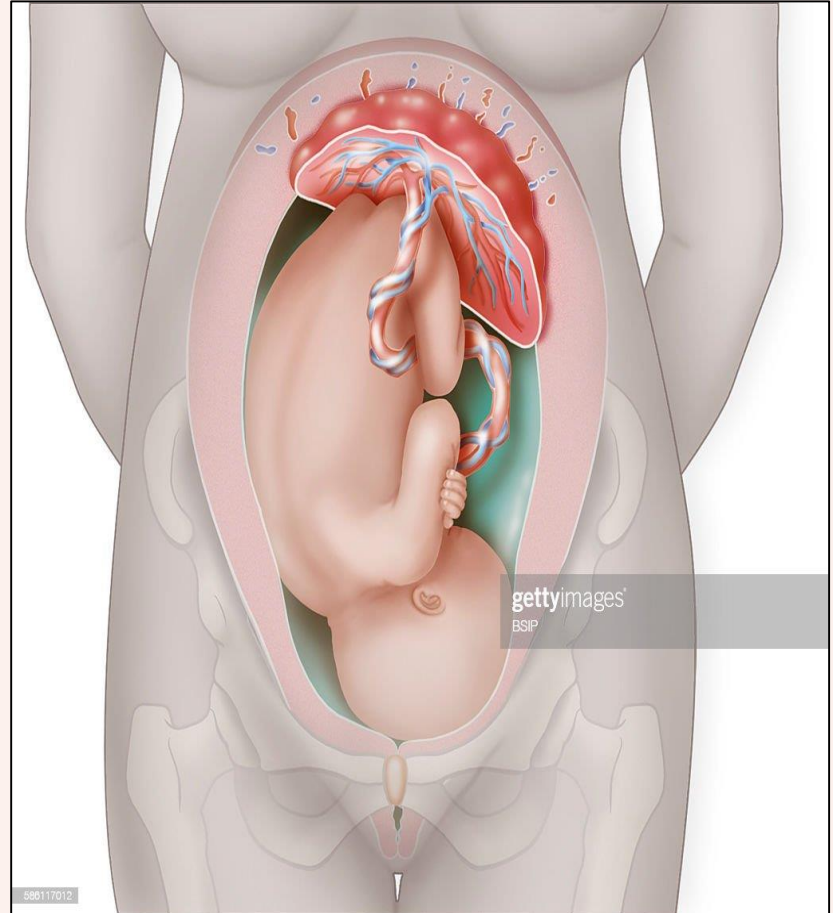
- Suspect when fetal descent is protracted or arrested.
- On vaginal examination
 - ⇒ Suture and fontanelles are palpable in the transverse diameter of the pelvis
 - ⇒ Fetal ears can be palpated superiorly under the symphysis and inferiorly above the sacrum/coccyx.

Management

- Expectant management:
 - ⇒ Small fetuses can usually be delivered
 - ⇒ If there is progress in descent and FHR is reassuring
 - ⇒ Partial or complete rotation may still occur spontaneously.
- Augmentation.
- Cesarean delivery(High transverse arrest,no maternal effort)

Fetal presentation

- Is the fetal body part which occupies the lower pole of the uterus and thereby first enters the pelvic passage
- The normal presentation is vertex .



Fetal presentation....

- Malpresentations

⇒ Are all fetal presentations other than vertex.



Face presentation



Brow presentation



Vertex presentation



Breech presentation



Shoulder presentation

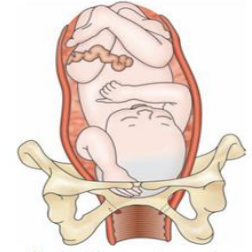
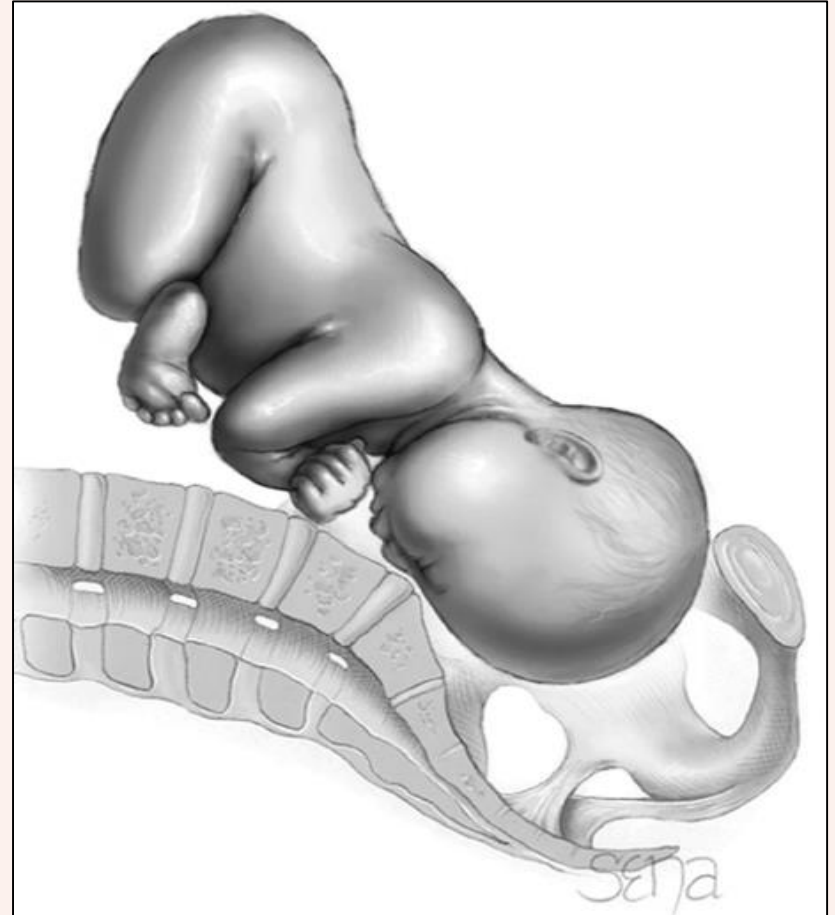


Fig. 1.4C: Compound presentation

Fig. 1.4A: Various types of fetal presentation

Brow presentation

- Partial extension of the fetal head making the occiput higher than the sinciput
- Presenting part are frontal bone, orbital ridges and anterior fontanel



Diagnosis

- Suggestive abdominal finding:
- Occiput felt above sinciput.
- On vaginal examination
- Anterior fontanelle and orbits are felt



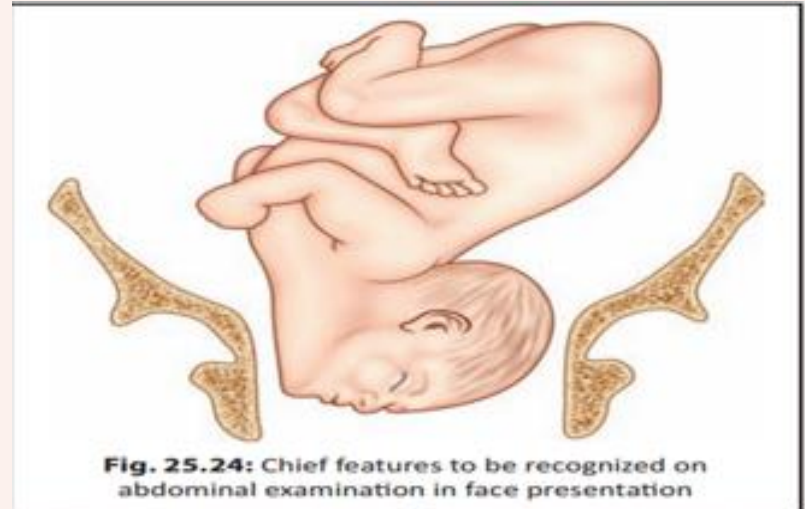
Management of persistent brow

- Persistent brow presentation is brow presentation in the later part of the first stage (after 6 cm of cervical dilatation) and second stages of labor.
- Deliver by cesarean section.



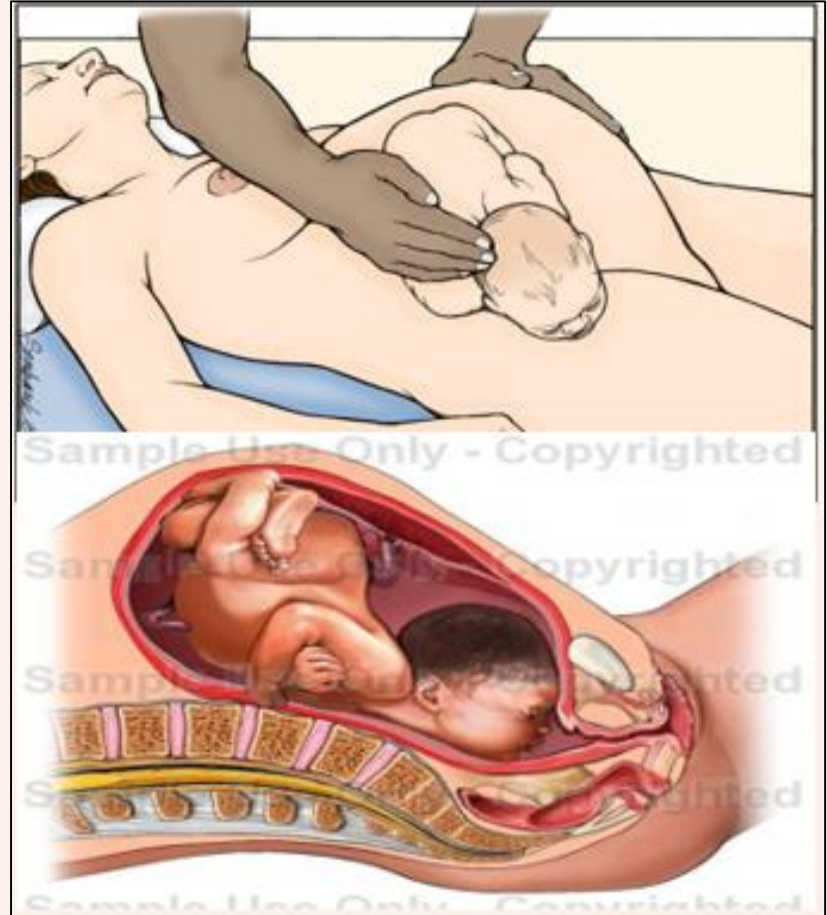
Face presentation

- Face presentation is hyperextension of the fetal head with the face being the leading part (figure 10).
- Fetal chin (mentum) is used as a reference point.



Diagnosis

- Suggestive abdominal finding
- ⇒ Groove may be felt between the occiput and the back (Leopold III).



Diagnosis...

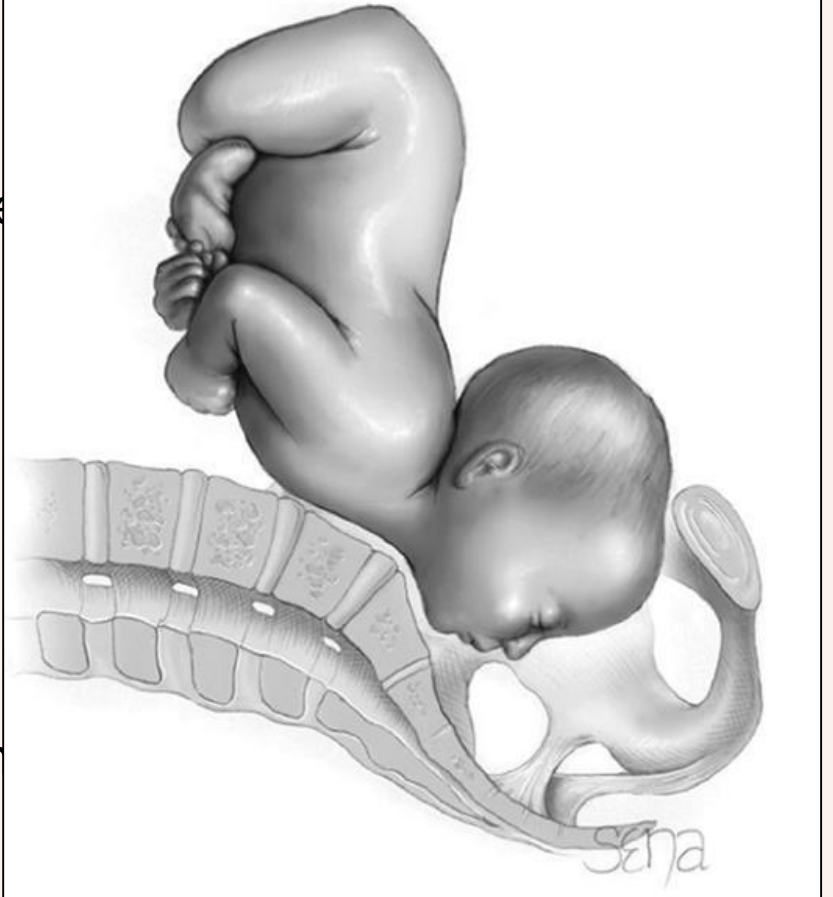
- On vaginal examination
 - ⇒ Fetal chin, mouth and nose palpated.
 - ⇒ The mouth with the two malar bone prominences make a triangle
 - ⇒ Mento-anterior: chin anterior position.
 - ⇒ Mento-posterior: chin posterior position



Fig. 25.25: Vaginal touch picture showing landmarks in mento-anterior position

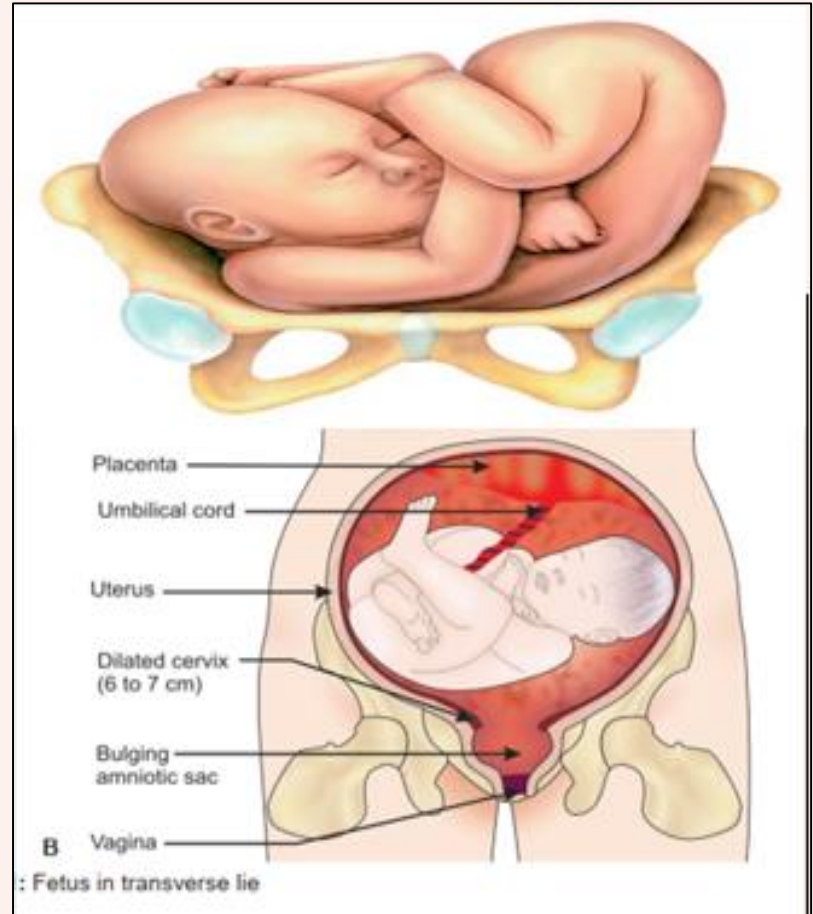
Management

- Mento-anterior:
 - ⇒ Adequate pelvis follow labor
 - ⇒ Forceps delivery can also be used
- Mento-posterior may rotate and follow labor
- Persistent mento-posterior
 - ⇒ If fetus is alive → CS
 - ⇒ If the fetus is dead → Craniotomy

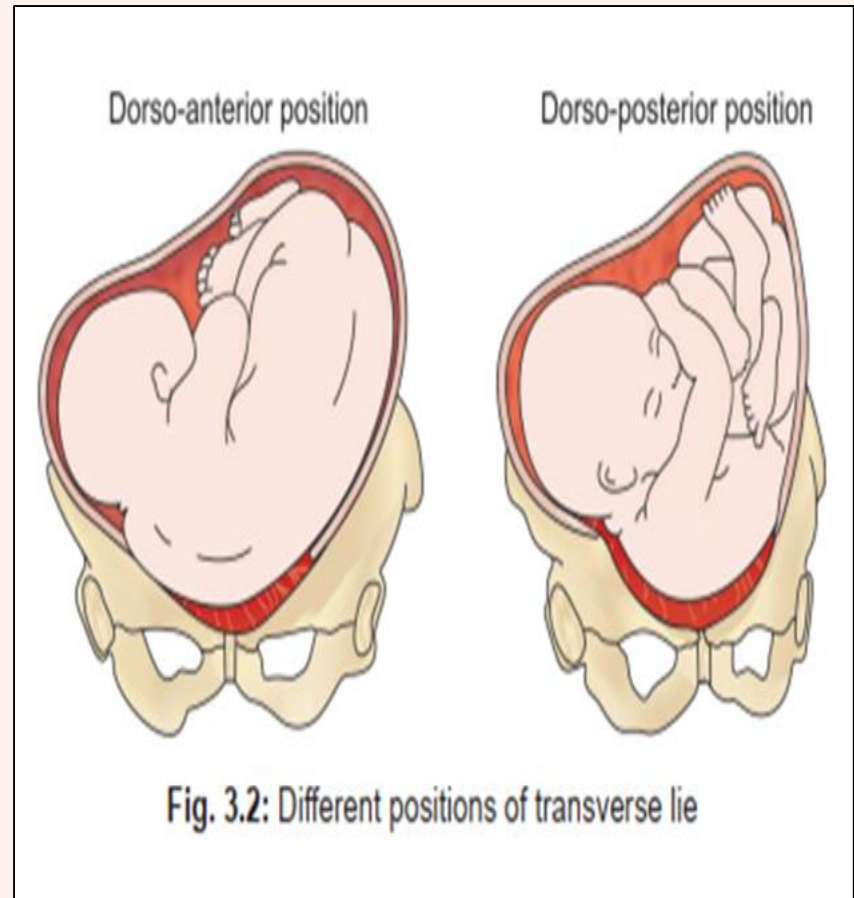
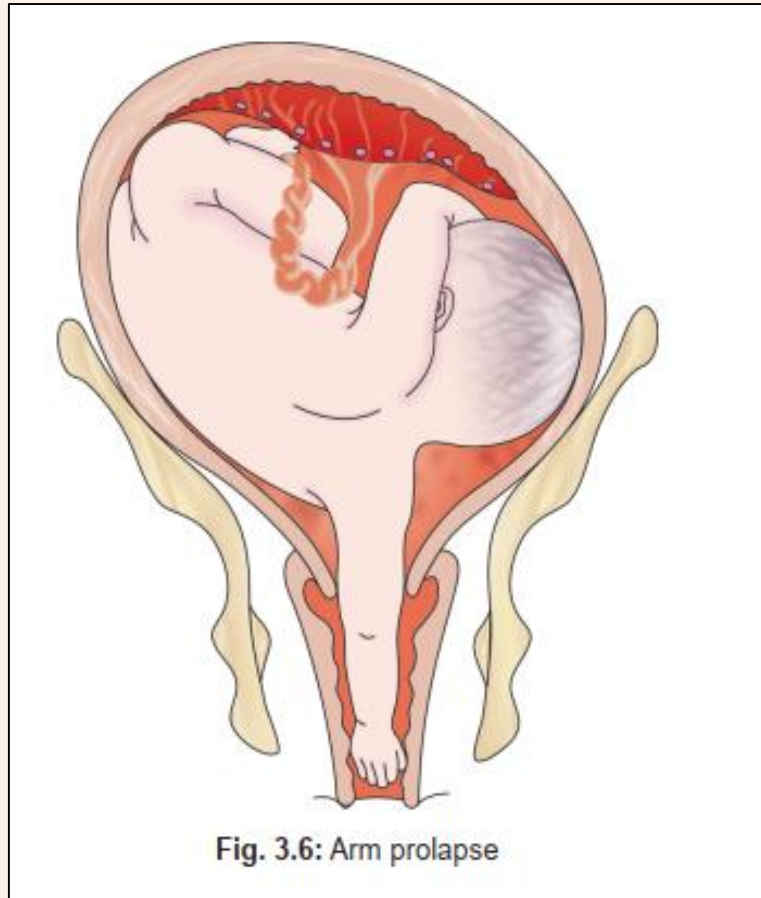


Transverse lie (shoulder presentation)

- Transverse lie is when the long axis of the fetus is perpendicular to the longitudinal axis of the mother/uterus.
- Shoulder presentation is when the shoulder is the presenting part in a transverse lie.



Transverse lie....



Diagnosis

- Abdominal findings:

- ⇒ Neither the fetal head nor the breech is felt in the upper and lower part of the uterus.
- ⇒ The abdomen is transversely elongated that longitudinally.
- ⇒ Fundal height is less than gestation age.



Diagnosis...

- Vaginal finding:
⇒ The shoulder or the prolapsed arm is felt.

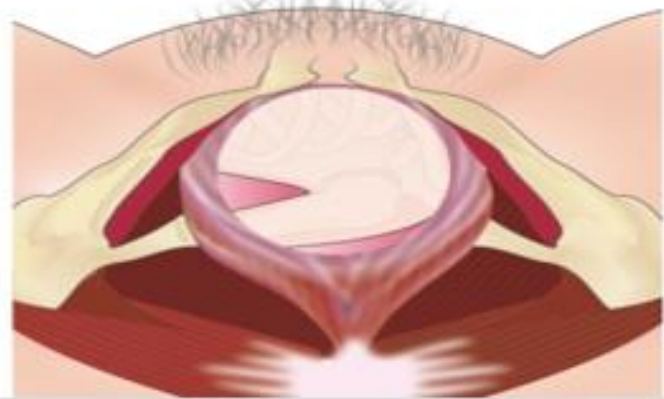
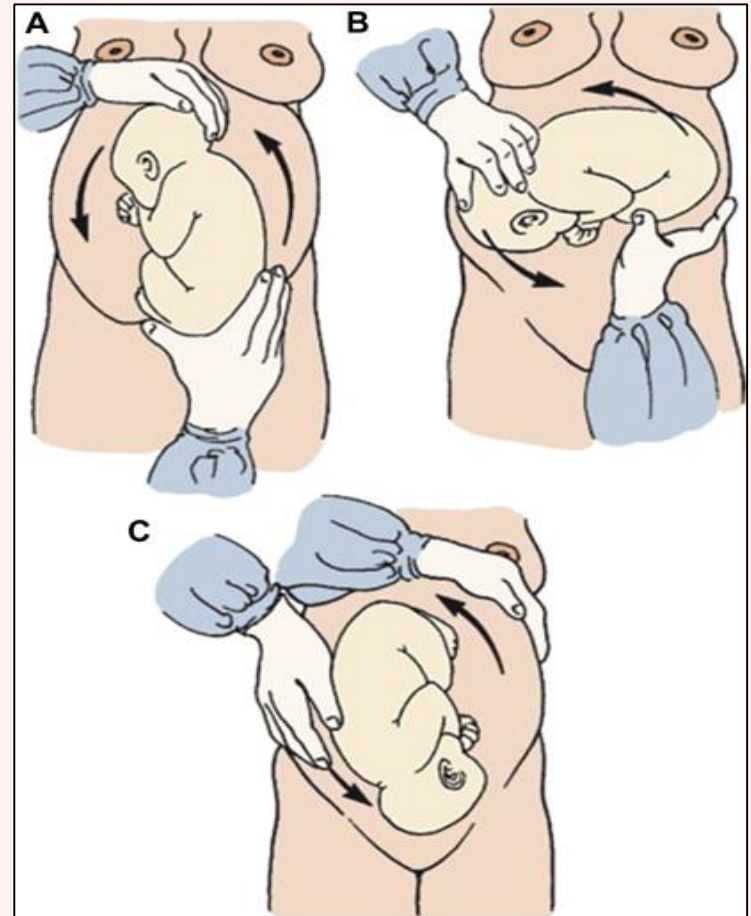


Fig. 25.32: Woman in labour with hand prolapse

Management

- External cephalic version
- Cesarean section



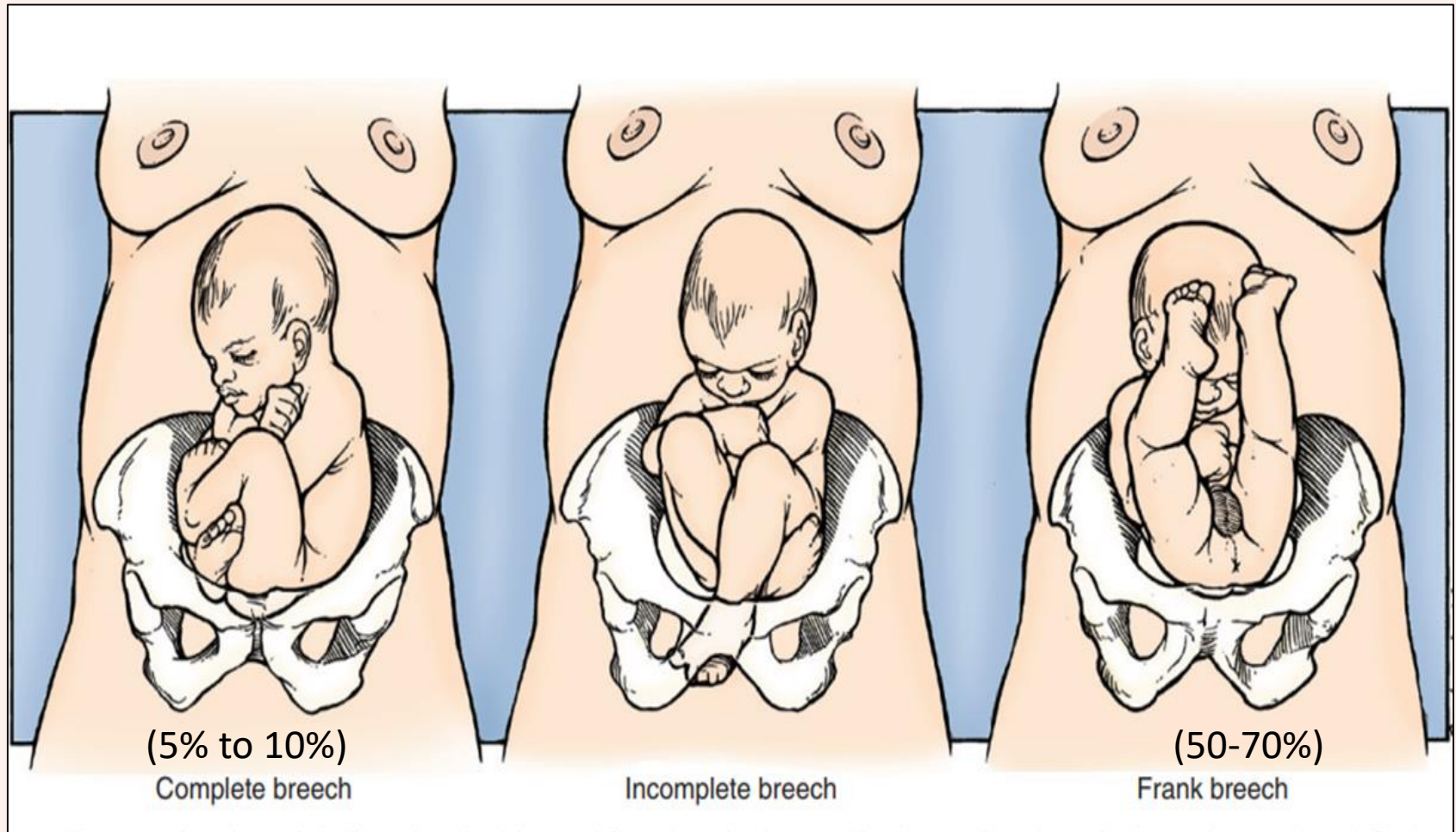
Breech presentation

- Is when the fetal buttock and/ or feet are the presenting part occupying the lower pole.
- Incidence
 - ⇒ 20% at 28th week
 - ⇒ 5% at 34th week
 - ⇒ 3–4% at term



Fig. 2.1: Breech presentation

Classification breech presentation



Predisposing factors

Maternal factor	Fetal Factors
• Cephalo-pelvic disproportion • Contracted maternal pelvis • Ppolyhydramnios, oligohydramnios) • Uterine anomalies (bicornuate or septate) • Placental abnormalities (placenta previa) • Multiparity (especially grand multiparas) • Previous history of breech delivery • Presence of pelvic tumor • Contracted pelvis.	• Prematurity • Fetal anomalies (e.g. neurological abnormalities,hydrocephalus, anencephaly and meningomyelocele) • Intrauterine fetal death

Diagnosis

- History
 - ⇒ Discomfort under the rib (due to the hard head)
 - ⇒ Feeling of more fetal movement in the lower segment



Diagnosis...

- Abdominal examination
 - ⇒ A round hard and smooth mass (head) at the fundus
 - ⇒ A soft, broad, and non ballotable mass (the breech) occupying lower pole of the uterus
 - ⇒ FHB loudest just above the umbilicus



Diagnosis...

- Vaginal examination
 - ⇒ Ischial tuberosities, sacrum and its spines, genitals and the anus



Diagnosis...

- Ultrasound
 - ⇒ Confirm clinical diagnosis
 - ⇒ Assess fetal attitude
 - ⇒ Assess amniotic fluid volume
 - ⇒ To estimate fetal weight, and to investigate for fetal anomalies and other predisposing factors



Management

- First stage of labor
 - ⇒ Secure IV line.
 - ⇒ Analgesics as labor pain management.
 - ⇒ Vaginal examination at rupture of membranes
 - ⇒ Avoid ARM.
 - ⇒ Instructed mother to push until the cervix is fully dilated
 - ⇒ Augmentation of labor is contraindicated

Management...

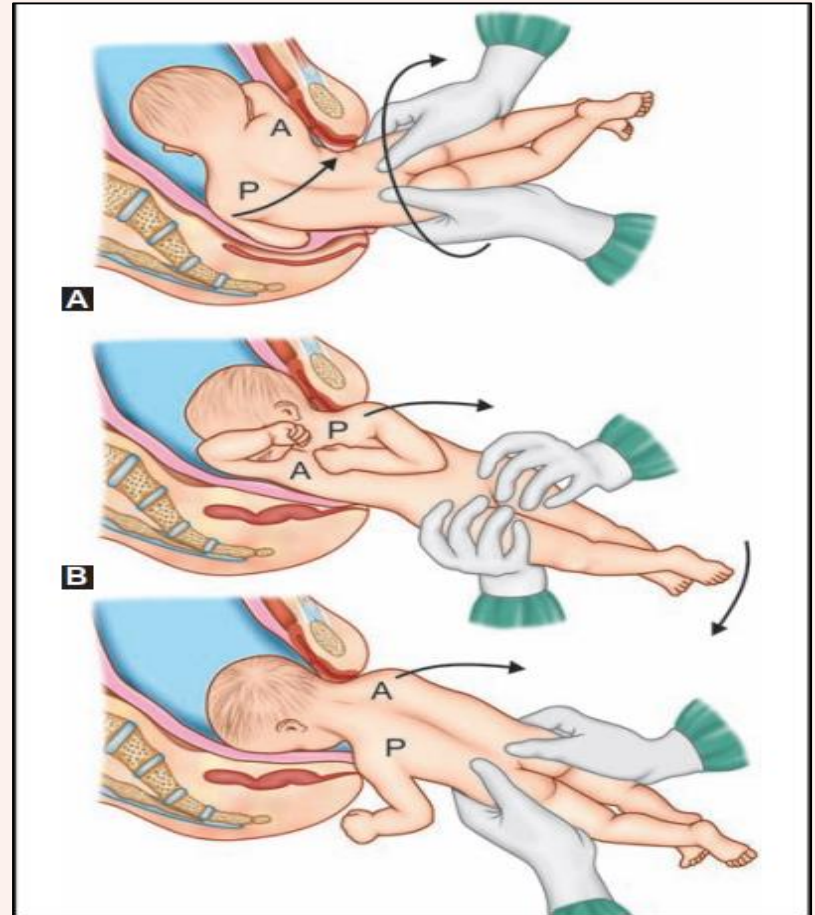
- Second stage of labor
 - ⇒ Wait buttocks till body is born to the level of the umbilicus
 - ⇒ If the legs do not deliver spontaneously
 - Pinard's maneuver



Management...

- Delivery of arm

⇒ Lovset's maneuver



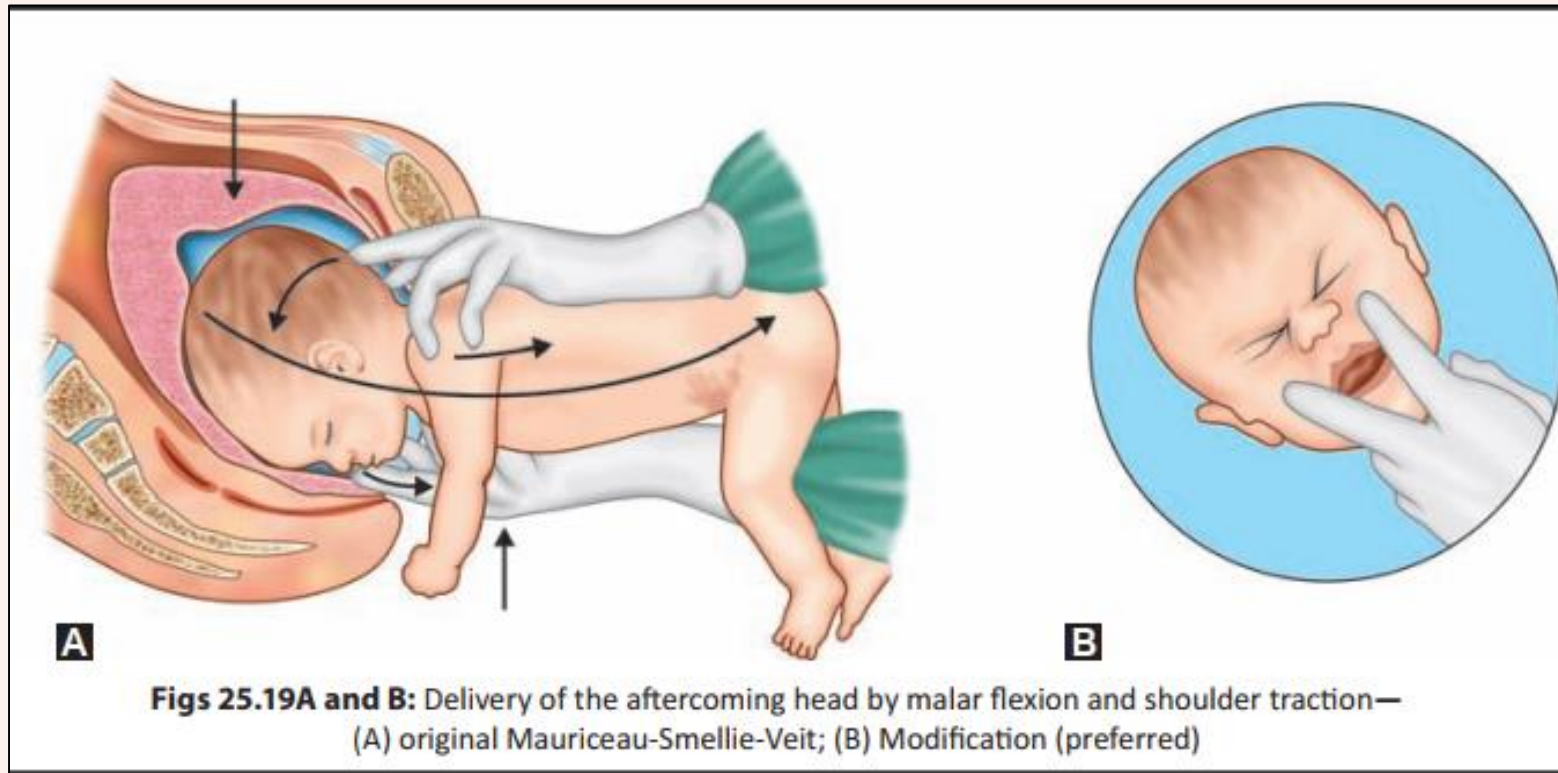
Management...



Figure 15. *Delivery of Arms and Shoulders.*

Management...

- Delivery of head :Mauriceau Smellie Veit Maneuver (MSV)



Management...

- Forceps delivery:
 - ⇒ Specialized type of obstetric forceps (Piper forceps) can be used to deliver after-coming head when the MSV cannot be accomplished easily.

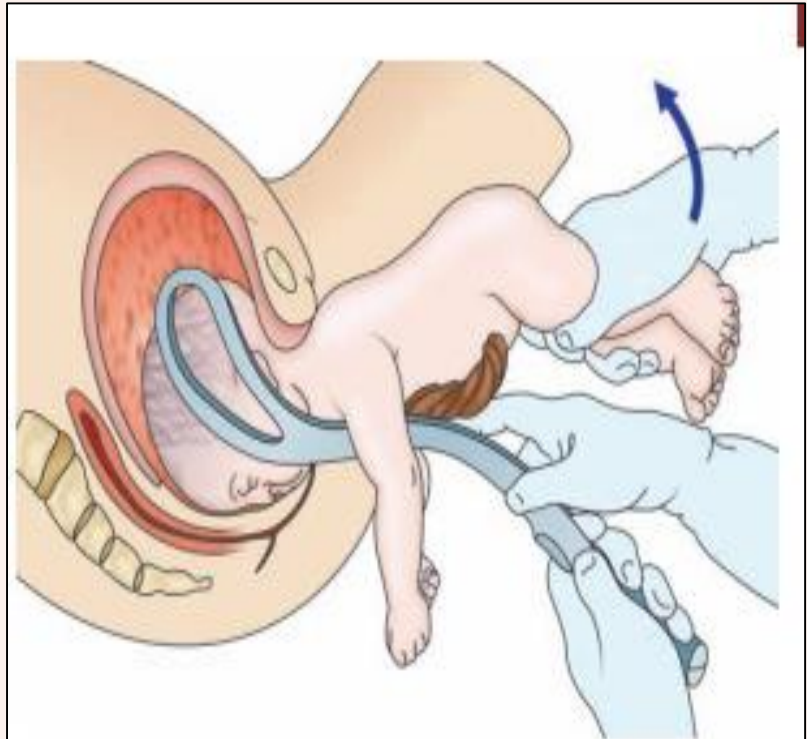


Fig. 2.10: Delivery of aftercoming head of the breech using forceps

Cephalopelvic disproportion (CPD)

- Is disparity between the fetal head and maternal pelvis which leads to inability of the fetal head to pass through the maternal pelvis for mechanical reasons.
- Failure of the fetal head to enter the pelvic inlet.



Fig. 23.9: Bimanual method of testing CPD. The finger tips are at the level of spines

CPD classification

1. Absolute

- The fetus is simply too large to negotiate the pelvis.
- True mechanical obstruction as a result of:
 - ⇒ Maternal(permanent)
 - ⇒ Temporary(fetal factors)

CPD classification...

Maternal(permanent)	Fetal(temporary)
• Contracted pelvis (commonest) (TC/OC/BTD < 8cm) • Pelvic exostoses • Spondylolisthesis • Anterior sacrococcygeal tumors	• Large fetus (macrosomia). • Shoulder dystocia. • Congenital abnormalities e.g., Hydrocephalus, ascites or tumor • Fetal monsters • Locked twins

CPD classification...

2.Relative

- Where the fetus may be delivered vaginally if a favorable combination of other factors can be achieved.
 - ⇒ Maternal-border line contracted pelvis (TC= 8-10cm).
 - ⇒ Fetal-mal presentation or position(Face,brow,OP).

Diagnosis

- Difficult before labor(ligaments & joints relax, fetal skull bones overlap)
- Trial of labor
- History
- Physical examination
- Radiological tests

Diagnosis....

- Antepartum examination, measurement and imaging cannot reliably diagnose CPD to preclude a trial of labor.

Diagnosis of CPD

History	Examination	Imaging
• Primigravid • Prolonged labor • Previous prolonged labor • Previous perinatal death • History of obstetric trauma • Properly documented obstetric record (e.g. intraoperative direct measurement of the obstetric conjugate)	• Arrest and protraction disorders of cervical dilatation. • Failure of head descent • High station of the head • Failure of progress of labor after correction of inadequate uterine activity (augmentation) • Fetal macrosomia • Fetal distress • Molding +2 and above • Caput • Dry vagina • Abnormal pelvimetry	• Ultrasound examination may reveal macrosomia or congenital anomalies e.g.hydrocephalus.

Diagnosis of CPD

True conjugate(TC)< 10 cm

Flat sacrum

Narrow sacrosciatic notch

Prominent ischial spine

Diverging/converging pelvic side walls

Bituberous diameter(BTD) < 8 cm

Narrow subpubic angle

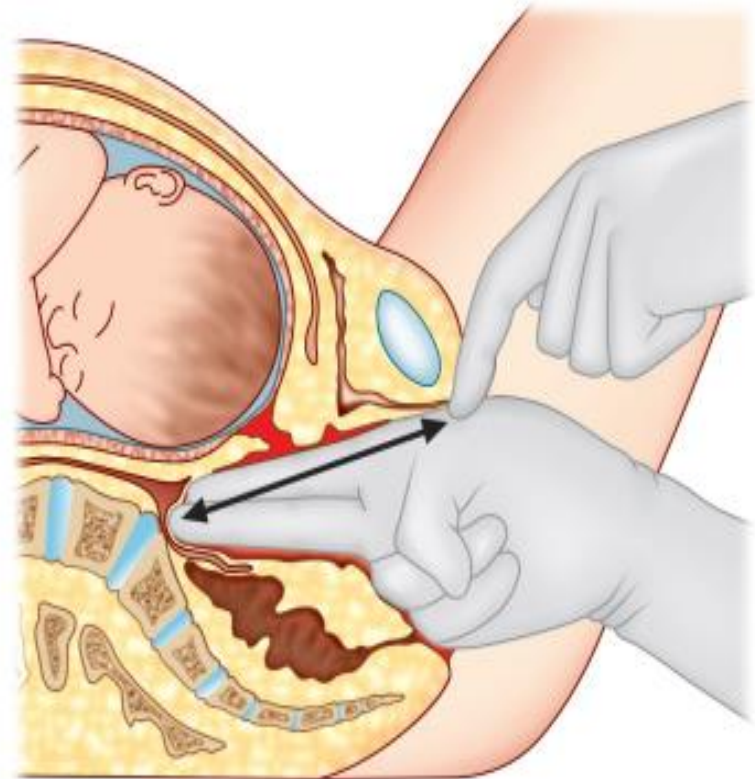


Fig. 9.10: Measurement of diagonal conjugate

Sign of contracted pelvis

Management of CPD...

- Route of delivery:
 - ⇒ Presence of CPD during labor is an indication for caesarean delivery.
 - ⇒ Craniotomy is indicated if the fetus is dead

Complications

Maternal	Fetal / neonatal
• Prolonged / obstructed labor: If CPD is not diagnosed & properly managed the end result is obstructed labor and its associated complications. • PPH • Maternal sepsis	• Fetal distress • Perinatal asphyxia • Neonatal infections • Perinatal death

Discharge counseling and education

- Explained about the indication (CPD) and the need for repeat CS in future.
- Feedback to referring health centers.
- Previous CS for CPD can be followed at a nearby health center and referred after 36 - 37 weeks of gestation.

Obstructed labor

- Is failure of descent of the fetus in the birth canal for mechanical reasons in spite of good uterine contraction.
- It is an outcome of neglected and mismanaged labor.

Cause of obstructed labor

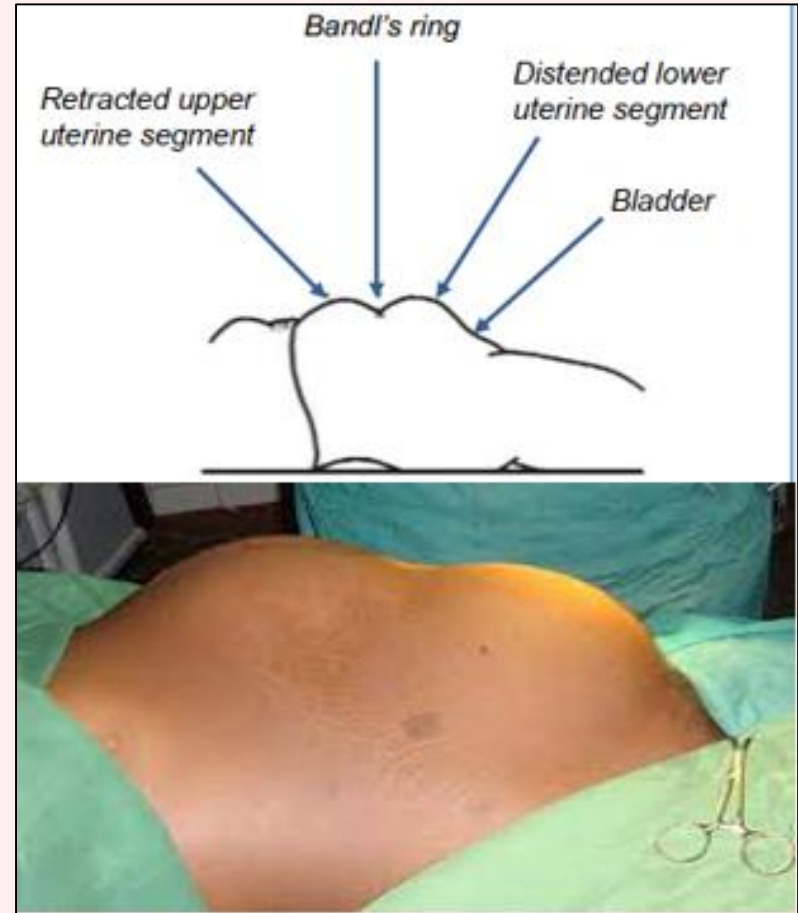
Maternal	Fetal
• Contracted pelvis / cephalopelvic disproportion (commonest • Soft tissue abnormalities ⇒ Tumor previa ⇒ Vaginal septum ⇒ Tight perineum ⇒ Uterine congenital anomalies	• Macrosomia • Malpresentations- e.g. shoulder • Malposition • Locked twins, conjoined twins • Fetal anomalies e.g. hydrocephalus • Shoulder dystocia

Clinical features

- History
 - ⇒ Abnormally prolonged labor
 - ⇒ Early ROM or prolonged rupture of the membranes
 - ⇒ Most do not have antenatal care
 - ⇒ Painful contractions
 - ⇒ Fever
 - ⇒ Previous prolonged labor, stillbirth or early neonatal death
 - ⇒ Previous operative deliveries
 - ⇒ Medical history;rickets, osteomalacia, or pelvic injury

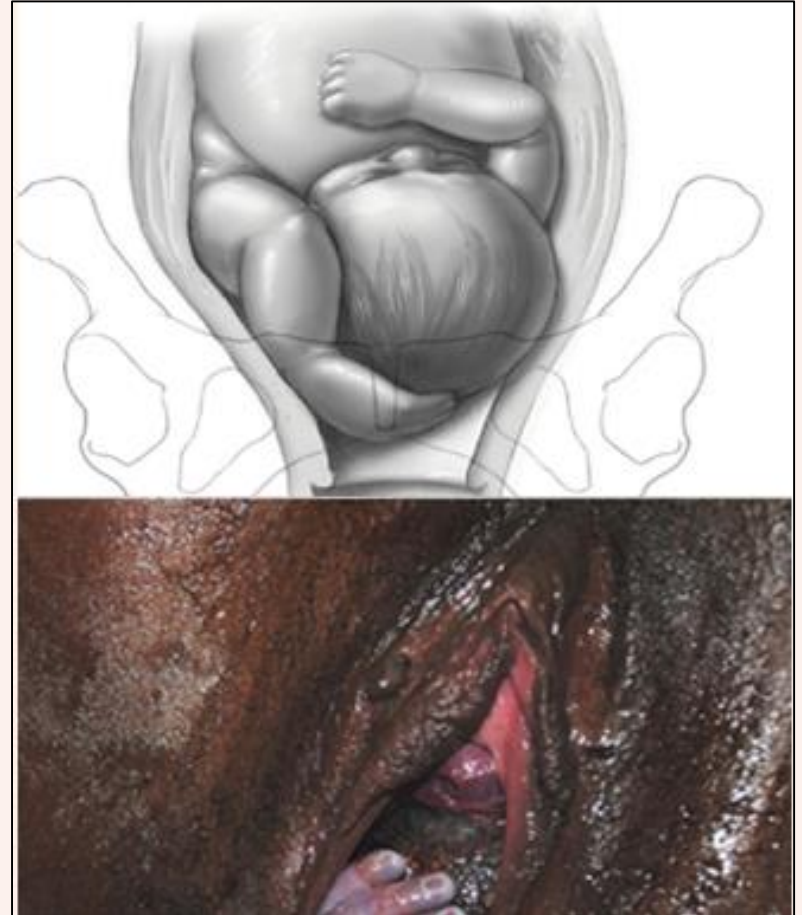
Clinical features...

- Abdominal findings
 - ⇒ Distention of the bowel as a result of acidosis and hypokalemia.
 - ⇒ Two/ Three tumor abdomen
 - ⇒ Abnormal FHB
 - ⇒ No fetal heart tone if the fetus dies from anoxia



Clinical features...

- Vaginal examination finding
 - ⇒ Edema of lower vagina and vulva (canula sign).
 - ⇒ Meconium- stained and foul-smelling discharge.
 - ⇒ Edema of the cervix
 - ⇒ Cervix poorly applied to the head.
 - ⇒ Little or no descent of the presenting part.
 - ⇒ Increasing molding & Caput



Management

- Secure good venous access
- Give crystalloids (ringers lactate is the fluid of choice)
- Give dextrose solution IV but never give orally.
- Monitor closely
- Blood group and Rh, cross-match, complete blood count
- Give broad spectrum antibiotics
- Catheterize
- Empty the stomach(NGT)
- Prophylactic dose of TAT: 1500 units
- Alive fetus: do emergency cesarean delivery.
- Dead fetus do destructive delivery if prerequisite fulfill

Complications

Early complications	Late complications:
• Atonic PPH • Uterine rupture • Peripartum infection • Tetanus • Maternal death • Fetal distress • Fetal & neonatal infections • Fetal and neonatal death.	• Fistula vesico-vaginal, rectovaginal) • Vaginal stenosis & stricture • Foot drop (sciatic, common peroneal nerve damage • Infertility following postpartum PID or hysterectomy • Psychological trauma due to the painful labor experience • Loss of the baby • Social isolation

Uterine rupture

- The complete nonsurgical disruption of all uterine layers.
- A large rupture:
 - ⇒ Massive hemorrhage.
 - ⇒ Extrusion of the fetus and/or placenta into the maternal abdomen.
- Small rupture may have minimal bleeding and insignificant maternal fetal consequences.

Uterine dehiscence

- An incomplete or occult uterine scar separation, in which the uterine serosa remains intact.
- Typically, no adverse obstetric outcomes are associated with a dehiscence.

Risk factor

- Multiple prior cesarean deliveries
- Induced or augmented labor
- Fetal macrosomia
- Postcesarean delivery infection
- Short interpregnancy
- Increasing maternal age
- Multiparity
- Fetal malpresentation
- Mid- to high-operative vaginal delivery
- Trauma

Risk factor...

TABLE 18-5

**RISK OF UTERINE RUPTURE BASED ON
INCISION OF PRIOR CESAREAN DELIVERY**

**INCISION OF PRIOR
CESAREAN DELIVERY**

**UTERINE RUPTURE
RISK (%)**

Classical	2-6
T or J shaped	2-6
Low vertical	2
Low transverse	0.5-1

Clinical manifestations

Symptom	Sign
• Vaginal bleeding • Cessation of fetal movement • Respiratory difficulty • Fever, chills, rigors • History of previous uterine surgery in some mothers	• Hypovolemia and shock • Acute cardio respiratory distress • Pallor • Dry mucosal surfaces • Tender abdomen • Easily palpable fetal parts • Fetal distress or death

Diagnosis

- Uterine rupture is suspected clinically
- Confirmed surgically.
- Ultrasound may confirm abnormal fetal position or hemoperitonium
- Laparotomy will demonstrate complete disruption of the uterine wall with hemoperitoneum and partial or complete extravasation of the fetus into the maternal abdomen

Management of uterine rupture

- Resuscitation
- Administer oxygen by face mask
- Open wide bore IV access
- Fluid resuscitation with crystalloids/colloids
- Administration of broad spectrum antibiotics
- Type and cross match blood for possible transfusion
- Catheterize bladder and monitor input-output
- Insert NG tube if food ingested recently
- Prepare for immediate laparotomy

Complications of uterine rupture

- Hypovolemic shock 20 hemorrhage
- Septic shock
- Stillbirth and early neonatal death
- Concomitant bladder injury
- Post partum anemia
- Maternal mortality
- Post operative infection
- Urethral damage
- Trombophlebitis
- DIC
- Hysterectomy and loss of fertility
- Pituitary failure & death

Prevention

- Most of the cause of uterine rupture can be avoided by good obstetric assessment & technique
- Supervised administration of oxytocin during labor.
- Estimation of fetal weight
- Use partograph

Premature rupture of membranes

- Membrane rupture that occurs spontaneously before the onset of labor
 - ⇒ Term PROM: Rupture of membranes after 37 weeks
 - ⇒ Preterm PROM: Before 37 weeks
- Prolonged PROM: Longer 12 hours
- Abc started after 8 hours if no labor started

Risk factors

- Mechanical factors: Multifetal gestation, polyhydramnios, pulmonary diseases, cervical conization/ LEEP/ cerclage.
- Urogenital infections: UTI, cervicitis, GBS, bacterial vaginosis.
- Previous history of preterm PROM, preterm labor.
- 2nd and 3rd trimester bleeding (e.g. abruption placenta).
- Other risk factors: Low socioeconomic status, nutritional deficiencies, low BMI, smoking and connective tissue disorders

PROM diagnosis

- History

- ⇒ Classic presentation is a sudden "gush" of clear or pale yellow vagina fluid.

- ⇒ Complications of PROM infection, preterm labor, cord prolapse

PROM diagnosis...

- Physical findings
 - ⇒ Negative discrepancy
 - ⇒ If complicated, uterine contraction, tenderness
 - ⇒ Speculum exam with or without Valsalva maneuver
 - ⇒ Best confirmatory methods direct observation of amniotic fluid

PROM diagnosis...

- Ferning test /arborized crystals
- Pad test
- Nitrazine paper test
- Dye amnioinfusion - golden
- Ultrasound: support diagnosis & fetal wellbeing.
- CBC
- U/A, culture & sensitivity

Complications

Maternal risks	• Infection of the amniotic cavity • Placental abruption • Higher risk of cesarean delivery
Fetal	• Infection • Fetal distress • Umbilical cord compression • Umbilical cord prolapse • Fetal death
Neonatal infection	• Acute congenital pneumonia • Sepsis • Meningitis • Late onset bacterial or fungal infections

Treatment

- Indications for expedite delivery
 - ⇒ Onset of labor
 - ⇒ Gestation age ≥ 37 wks
 - ⇒ Evidence for nonreassuring fetal status
 - ⇒ Evidence for chorioamnionitis
 - ⇒ Lethal congenital anomalies
 - ⇒ Intrauterine fetal death
 - ⇒ High risk of cord prolapse and abruption placenta
- Expectant management for gestational is below 34 weeks and both the fetal and maternal conditions are stable.

Treatment

- Expectant management

- ⇒ Admit to the ward

- ⇒ Avoid digital cervical (pelvic) examination.

- ⇒ Advise bed-rest

- ⇒ Dexamethasone 6 mg IM BID for 48hrs for lung maturity

- ⇒ Antibiotics

1. Ampicillin 2gm IV QID and Erythromycin 250 mg P.O QID for 48 hours then
2. Amoxicillin 500 mg P.O TID & Erythromycin 250 mg. P.O QID for 5 days